

Causal-Order Advantage Estimation: Exploiting Token Lineage for Credit Assignment in Reinforcement Learning for Dynamic Task Assignment

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Abstract

Deep reinforcement learning is increasingly applied to dynamic task assignment: routing stochastic streams of work to limited, contended resources. When such a system is modelled as an Action-Evolution Petri Net, the simulator records, at no extra cost, the *provenance* of every token—which past decisions causally influenced which rewards. Existing methods discard this and hand the net to off-the-shelf policy-gradient learning as a black-box semi-Markov decision process. Generalized Advantage Estimation then accumulates temporal-difference errors along the *trajectory index*, but in a system running many concurrent cases the next decision usually belongs to a different case, so its error term is largely noise with respect to the current action. We run the identical recursion along the provenance DAG instead. The resulting estimator assigns each reward to a single owner—making credit a return decomposition rather than a hindsight Q-sample—and combines causal successors by a flow-weighted average normalized over successors. We show that the natural formulation, which normalizes over *predecessors*, leaves the bootstrap coefficient unbounded and consequently *diverges* during training; bounding that coefficient is what is necessary, and normalizing over successors is the canonical way to achieve it. On five problem configurations spanning one to eight independent reward-bearing components, causal-order estimation improves normalized return over discount-matched PPO by +0.19 to +0.51 (paired, 20 seeds, $p \leq 0.002$) on the three most decomposable of them, and is statistically indistinguishable from it on the two least, all while costing about one millisecond per episode. We further show that a structural statistic computed before training—the number of reward-bearing causal components—predicts where the method helps and where it provably cannot, and we report a battery of falsified alternative explanations so that the mechanism claim is auditable rather than asserted.

Keywords: reinforcement learning, credit assignment, dynamic task assignment, Petri nets, decomposition, policy gradient

1. Introduction

Dynamic task assignment—deciding, repeatedly and under uncertainty, which resource should serve which unit of work—is among the most widely studied operational control problems in operational research, covering machine scheduling, service operations, patient flow, order picking and vehicle dispatching. Recent work models such problems as Action-Evolution Petri Nets (A-E PNs) and solves them with deep reinforcement learning, which handles the

combinatorial, variable-dimension action spaces that classical policies struggle with.

That line of work treats the learner as interchangeable. The net is compiled into a semi-Markov decision process and handed to proximal policy optimization with Generalized Advantage Estimation (GAE). In doing so it discards something the formalism provides for free.

The credit-assignment problem in concurrent systems.. GAE accumulates temporal-difference errors along the trajectory index, $A_t = \delta_t + \gamma \lambda A_{t+1}$. In an A-E PN running many cases concurrently, decision $t + 1$ typically belongs to a *different case* from decision t : its reward, and hence its TD error, is driven by tokens the action at t never touched. Credit therefore propagates along wall-clock order while causality flows along the provenance DAG. The consequence is variance that grows with concurrency, and it is severe. At the converged training budget on our scaling benchmark with four independent copies, discount-matched PPO reaches normalized return 0.715 ± 0.265 and collapses on one seed in twenty, where the estimator proposed here reaches 0.980 ± 0.092 and collapses on none. On the most decomposable environment the same contrast is 0.161 against 0.673, with PPO collapsing on 14 seeds of 20 and the proposed estimator on none.

What the formalism already records.. An A-E PN simulation logs, for every token, which transition firing produced it from which input tokens. This provenance DAG is not an addition to the model; it is a by-product of executing it. It answers exactly the question credit assignment asks: which earlier decisions could have influenced this reward?

Contributions..

1. **Causal-order advantage estimation** (CGAE-CF, Section 4): GAE with the temporal successor replaced by the *causal* successor, with single-owner reward attribution and a convex flow-weighted combination over successors. The credit factorization is *derived* from provenance per episode rather than hand-specified, requires no forked simulation, and costs about one millisecond per episode—under 8% of a training iteration, in which the graph neural network dominates.
2. **Identification and repair of a normalization defect** (Section 5). The natural flow-weighted form normalizes weights over the *predecessors* of each node, which is what a token-conservation argument yields. But the recursion sums over *successors*, so the coefficient multiplying the bootstrap is a row sum that the conservation constraint does not bound. We show the resulting error is proportional to the critic, so the estimator degrades as the critic improves, and we observe exactly that: on the single-component environment the unnormalized form does not merely underperform, it *diverges*, with variance growing sixfold between training budgets and one run falling from near-optimal to below random.
3. **An ablation ladder in which computing the estimand *exactly* does not pay** (Section 6). CGAE-DAG computes the causal estimand by construction and ranks second-worst of nine, beaten decisively by three biased variants; the one variant that fails catastrophically is the one whose error is *unbounded*. What separates working

from failing estimators here is boundedness, not exactness—a distinction we can support, in contrast to several mechanisms we tested and rejected (Appendix Appendix D).

4. **A pre-training diagnostic.** The number of reward-bearing causal components K , and the mean causal fan-out, are computable from a handful of random rollouts. Together they predict both whether the method can help ($K > 1$) and whether the choice of successor-combination rule matters at all (fan-out > 1). We report a case where fan-out is exactly 1 and every variant is provably bit-identical.
5. **A boundary characterization** (Section 7, carried over): three independent routes to supplying the learner with more structure — potential-based reward shaping, structural input features, and a graph-actor architectural fix — each provably inert or empirically neutral on a single-component bottleneck.

Scope. We deliberately do not claim superiority over prior component-factored credit schemes. At converged training budgets *cgae-cf* is statistically indistinguishable from *ccf* and from the mean-combination variant (Section 6); the differences reported in earlier drafts of this work were artifacts of a shorter training budget. What is established is that causal-order estimation beats discount-matched PPO decisively where causal structure exists, and that the successor normalization is necessary to avoid divergence.

2. Related work

DRL for dynamic task assignment and operational control. A growing literature applies DRL to scheduling, routing, and business-process decision-making, typically encoding the problem instance as a graph consumed by a graph neural network policy: greedy graph-embedding heuristics for combinatorial construction (Dai et al., 2017), attention-based routing policies (Nazari et al., 2018), and GNN-based dispatching for job-shop scheduling (Zhang et al., 2020), the last especially close in spirit to our setting. Mazyavkina et al. (2021) survey this broader RL-for- combinatorial-optimization literature; Bengio et al. (2021), published in this journal, give the methodological tour d’horizon our OR framing draws on most directly. Within the A-E PN program, Lo Bianco et al. (2023) introduce the formalism and a taxonomy of assignment archetypes, Lo Bianco et al. (2025b) give a graph representation for infinite state/action spaces, and Lo Bianco et al. (2025a) provide a library supporting partial observability and multiple decisions. Across this literature, the learning algorithm is almost universally a black box on top of the graph encoding; the simulator’s own execution trace—which decisions caused which outcomes—is not itself treated as a source of learning signal. Our contribution is orthogonal to and compatible with all of the above: it is a credit-assignment mechanism, not a new policy architecture or problem encoding.

Credit assignment in RL. Return decomposition (Arjona-Medina et al., 2019) learns to redistribute return along a trajectory via a return-predicting model’s contribution analysis; Hindsight Credit Assignment (Harutyunyan et al., 2019) reweights past actions by their learned likelihood of having caused an observed outcome. Both are *learned* redistribution mechanisms trained on correlational patterns across trajectories. Our mechanisms differ in kind: token

provenance in an A-E PN is not inferred, it is recorded exactly by the simulator, so CGAE-CF’s credit signal carries no approximation error from the attribution step itself. This is close in motivation to model-free counterfactual credit assignment (Mesnard et al., 2021), which conditions value functions on future events to build low-variance counterfactual baselines; Simulator-replay counterfactual credit (CFPK, which we treat as related rather than contributed work in this version) is the model-based analogue of the same goal, made exact because the simulator is itself an available, cheap-to-fork causal model. It sits at the exact end of the same spectrum, at a simulation cost CGAE-CF avoids entirely.

Factored / multi-agent credit.. Difference rewards (Wolpert and Tumer, 2001) credit each agent with its marginal contribution relative to a counterfactual baseline; COMA (Foerster et al., 2018) learns this baseline via a centralized critic; value decomposition (Sunehag et al., 2018; Rashid et al., 2018) factors a joint value across agents under additivity or monotonicity constraints. These require the entity/agent partition to be given. A more recent line does not: MACCA (Ruan et al., 2024) casts the generative process as a dynamic Bayesian network and *learns* the causal structure from offline data, using a structure predictor with sparsity regularization to recover which state and action dimensions influence each agent’s reward.

The distinction we draw is therefore not *given versus derived*—that would misrepresent Ruan et al. (2024)—but *learned versus recorded*. MACCA estimates the causal structure statistically from $\{s, o, a, R, s'\}$ tuples, so the recovered graph carries estimation error, requires enough data to identify, and can be wrong. Token provenance in an A-E PN is not estimated at all: the simulator writes down, for each firing, exactly which tokens were consumed and produced, so the attribution step contributes *zero* approximation error and needs no data beyond the episode being credited. The partition is also free to change shape per episode as the realized case mix changes, and a single-agent problem (one policy, many concurrent cases) is factored the same way a genuinely multi-agent one would be.

Treating the simulator as more than a black box.. Two recent lines share our premise—that a discrete-event simulator exposes structure worth exploiting—while exploiting different structure. Che et al. (2024) make the simulator *differentiable* and compute pathwise policy gradients through it for queueing-network control, reporting gradient estimates orders of magnitude more accurate than REINFORCE-style estimators and large sample-efficiency gains. That approach needs a differentiable simulator and smooth surrogate event dynamics; ours needs only that the simulator record token identity, which A-E PN execution already does, and leaves the policy-gradient estimator otherwise untouched. The two are complementary rather than competing: pathwise differentiation improves the gradient of a given objective, whereas causal-order estimation changes which rewards enter a decision’s advantage in the first place. Yeganeh et al. (2026) control a semiconductor fab—an event-driven system much like ours, at industrial scale—and begin from an observation we share almost exactly: because an action initiates a temporally extended event that overlaps other events, “the next decision point may correspond to a different event and thus be uninformative for credit assignment.” We record this agreement explicitly, because it is independent evidence that the defect our method targets is real and is felt in practice.

Their response to it differs from ours in the variable it groups on. They define transitions over the temporal extent of the action-induced event rather than to the next arbitrary decision—an SMDP-style retiming, as they note—and then add an *event-group* temporal-difference layer: all events falling in a sampled time segment $I_h = [t, t + h]$ form a group G , and training matches the group’s aggregated TD discrepancy to a single system-level reward earned over that interval, $\mathcal{L}(\theta) = (|G|^{-1} \sum_{e \in G} \delta_e(\theta) - r^g(I_h))^2$. A weighted variant replaces the mean with normalized *duration* shares $\omega_e \propto \Delta_e$ over the events overlapping a sampled event’s interval. For policy optimization they set GAE aside as ill-suited to very long delays and use a group-relative surrogate $\tilde{A}_i \triangleq \delta_i - r^g$, which subtracts the shared broadcast reward from the critic-implied return in the manner of a group baseline.

So their layer does redistribute a delayed global reward across decisions, and it would be wrong to call it only a retiming. But the group is a set of events that *co-occur in time*, and the weights are shares of elapsed duration. Ours is the set of decisions a reward causally *descends from*, and the weights are shares of token flow. The two coincide only by accident: two lots processed in the same 360-minute window are grouped together there whatever their provenance, and two decisions on one wafer’s route are separated there once they fall in different windows, while our partition does the opposite in both cases. The consequences follow from that difference. Their aggregation needs a system-level reward signal defined over an interval to aggregate *toward*; ours needs only the rewards the episode already pays, each assigned to a single owner, which is what makes it a return decomposition rather than a reward-matching objective. Theirs applies to any discrete-event simulator that reports event start and finish times; ours additionally requires that the simulator record token identity, which is a stronger demand that A-E PN execution happens to satisfy for free. And where they identify the theoretical status of their objective as open—“whether and when it admits, approximates, or preserves a Bellman-consistent fixed point”—the corresponding question for our estimator is what Section 5 answers, though for a weaker property: a bound on the bias relative to the closure estimand rather than a fixed-point guarantee.

Potential-based reward shaping. Ng et al. (1999) prove that adding a potential-based term to every step’s reward cannot change an MDP’s optimal policy, for any potential function. We use the SMDP-discounted generalization of this theorem not as a headline mechanism but as one of three falsified alternative routes in our boundary-condition investigation (Section 7): a theorem-safe, empirically counter-productive result at practical training budgets, included because it isolates sample-efficiency effects cleanly from the bias effects that are our main concern in Section 5.

Graph neural networks for combinatorial optimization and heterogeneous actor-critic architectures. GNNs are now a standard state encoder for RL over combinatorial structures with variable size and topology (Dai et al., 2017; Zhang et al., 2020); our environments use a heterogeneous variant, the Heterogeneous Graph Transformer (Hu et al., 2020), encoding places and transitions as distinct node types. Section 7 documents an architectural finding of independent interest: a per-node actor decoder combined with directed-only (non-reversed) message-passing edges can be *exactly* blind to state outside its forward-reachable neighborhood, a limitation the critic (which pools) does not share. To our knowledge this is not previously documented for heterogeneous actor-critic graph architectures, and the argument is orthogonal to the specific choice of GNN backbone.

Decomposition in operations research.. Dantzig–Wolfe decomposition (Dantzig and Wolfe, 1960) and Benders decomposition (Benders, 1962) break a hard, structured optimization problem into a master problem and weakly-coupled subproblems along a hand-identified block structure. CGAE-CF applies the same idea to the policy-gradient *return*: decompose it along a block structure derived automatically, per episode, from the simulator’s own causal record, rather than a static constraint matrix’s block structure identified a priori.

Generalized advantage estimation, and deliberate bias.. Our estimator is GAE with one substitution, so GAE is the closest technical antecedent rather than background. Two features of it matter here. First, GAE with $\lambda < 1$ is *itself* a biased estimator— $\lambda = 1$ is the unbiased Monte-Carlo case—and the biased settings are universally preferred in practice. Second, the bias it introduces is a shrinkage by *temporal* distance, λ^k ; the estimator below shrinks by *causal* distance instead. The precedent is therefore direct: we are not introducing bias into an unbiased pipeline, we are relocating a shrinkage that is already there.

Biased estimators as the norm.. It is worth stating plainly that deliberate bias is standard rather than exceptional in this literature: the clipped surrogate objective of PPO, truncated importance weights in off-policy actor-critic, reward clipping in value-based control, restricted value factorizations in cooperative multi-agent learning, and discounting below the true horizon as a form of regularization. In statistics the same point is a theorem: shrinkage estimators dominate the unbiased maximum likelihood estimator in mean-squared error. We rely on this framing in Section 5, where the estimator we ship is biased and the one that is exactly unbiased performs worse.

Positioning against factored and counterfactual credit.. Difference rewards, counterfactual multi-agent baselines and value-decomposition networks all require the entity partition to be supplied. Our factorization is *read off* the provenance DAG per episode. Simulator-replay counterfactual credit obtains exact per-decision differences by forking the simulator; it sits at the exact end of the same spectrum, at a cost in simulation we avoid.

3. Background

Because Action-Evolution Petri Nets are not yet widely known, we describe the formalism intuitively before defining the learning problem.

Petri nets and colored tokens.. A Petri net is a bipartite directed graph of *places* (drawn as circles) and *transitions* (drawn as bars), connected by arcs. Places hold *tokens*; in a *colored* net each token carries typed data—a task’s type, a resource’s identity. A transition is *enabled* when its input places jointly hold a matching combination of tokens, called a *binding*; *firing* the transition consumes that binding and produces new tokens in its output places. The distribution of tokens over places—the *marking*—is the system state, and it evolves as transitions fire.

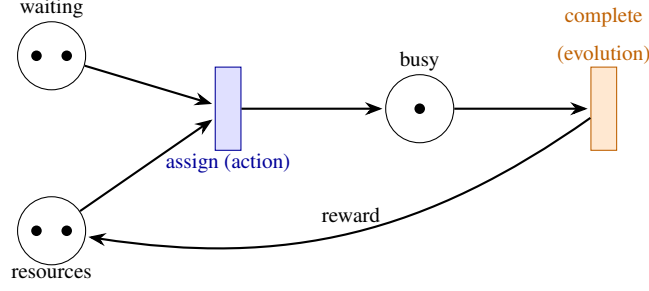


Figure 1: A minimal A-E PN for task assignment. Circles are places holding colored tokens; bars are transitions. The *assign* **action** transition (blue) is a decision point: the policy chooses which waiting task to bind to which free resource. The *complete* **evolution** transition (orange) fires automatically after a stochastic delay, returns the resource, and emits a reward.

The action/evolution split. An Action-Evolution Petri Net (A-E PN) (Lo Bianco et al., 2023) partitions the transitions into two kinds and adds a global clock. *Evolution* transitions model the exogenous dynamics: they fire automatically according to stochastic, timed rules—task arrivals, service completions—advancing the clock. *Action* transitions model the controllable decisions: whenever several action bindings are simultaneously enabled, an external *policy* selects which one to fire. Selected firings may carry a scalar *reward*. Figure 1 shows a minimal assignment net: a waiting task and a free resource are consumed by an *assign* action transition—the decision of which task to serve with which resource—producing a *busy* token; an evolution transition *complete* later frees the resource (returning it to the pool) and emits a reward.

The induced SMDP. Executing an A-E PN therefore alternates between the environment evolving the marking (evolution firings that advance the clock) and the agent choosing among enabled action bindings—exactly a semi-Markov decision process (SMDP). We write s_d for the marking at decision d , a_d for the chosen binding, and u_d for its clock; delayed rewards are discounted by the SMDP factor $\rho_d(t) = e^{-\beta \max(0, t-u_d)}$. Dynamic task assignment problems—matching stochastic task streams to contended resources—map onto this template directly (Lo Bianco et al., 2023, 2025b,a). Because the simulator is a deterministic function of the marking, the clock, and the (recorded) exogenous random draws, it can be *snapshotted and re-run* from any decision point under an alternative action—the capability CFPK’s forking (Section 4) relies on, and a further consequence of the same executable-model property that makes A-E PN a unified modeling-and-solving framework in the first place.

Policy gradient over A-E PN. The policy π_θ is a graph neural network over the assignment-graph representation (Lo Bianco et al., 2025b), trained by PPO (Schulman et al., 2017) with generalized advantage estimation (Schulman et al., 2016). The canonical unbiased gradient estimator credits each decision with its reward-to-go against a learned baseline $V(s_d)$; both mechanisms below modify only how that reward-to-go is computed.

Provenance DAG. Executing an A-E PN produces a record, for each firing, of the input tokens consumed, the output tokens produced, the firing time, and any reward emitted. Writing $\mathcal{A}$ for the set of *agent decisions* in an episode,

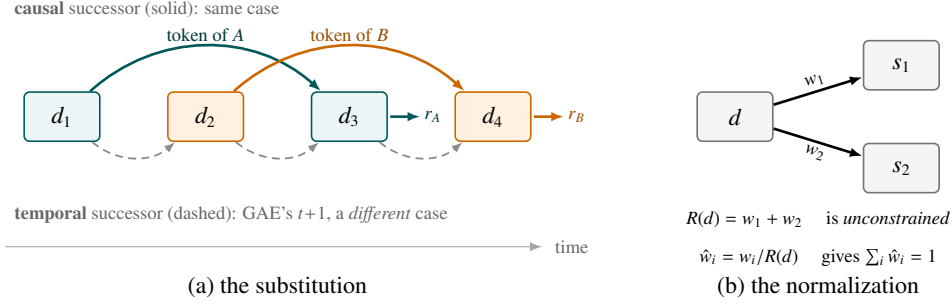


Figure 2: The mechanism. **(a)** Four decisions in wall-clock order, drawn from two interleaved cases A and B . Generalized advantage estimation bootstraps along the dashed *temporal* chain, so the TD error it charges to d_1 is the one produced at d_2 —a decision about a different case, whose reward d_1 did not influence. The token d_1 produced is instead consumed at d_3 , where the reward r_A is emitted. Causal-order estimation follows the solid provenance edges, so d_1 is credited with r_A and never with r_B . Reward ownership is a partition—each reward is assigned to the latest decision on its own lineage—so no reward is counted twice. **(b)** Where a decision’s tokens feed several later decisions, the edge weights must be normalized *over successors*. Token conservation constrains the sum over each node’s predecessors, which leaves the row sum $R(d)$ free; because $R(d)$ multiplies the bootstrapped value (Proposition 2), leaving it unbounded makes the recursion diverge in training.

indexed in firing order with clock values u_d , the token history induces a directed acyclic graph on $\mathcal{A}$: token identity links each consumed token back to the firing that produced it, transitively through non-decision (forced and evolution) firings.

Generalized advantage estimation.. For completeness, GAE forms $A_t = \sum_{k \geq 0} (\gamma \lambda)^k \delta_{t+k}$ with $\delta_t = r_t + \gamma V(s_{t+1}) - V(s_t)$, computed by the backward recursion $A_t = \delta_t + \gamma \lambda A_{t+1}$. The substitution in Section 4 replaces the successor $t+1$ in this recursion.

4. Method: causal-order advantage estimation

4.1. Causal successors

Figure 2 shows the construction on a short episode: the trajectory index along which GAE ordinarily accumulates, and the provenance DAG that replaces it. For a decision $d \in \mathcal{A}$, define $s \in \text{succ}(d)$ when s consumes a token descending from an output token of d with no intervening decision—the *nearest producer* relation. Forced firings and evolution transitions are transparent: the backward walk passes through them to the decision that actually caused the flow. This is a purely structural construction requiring no token-value semantics and no assumption about what the tokens represent.

Remark 1 (A known specification wrinkle, disclosed). *Under token-flow postponement, a POSTPONE decision re-emits the entire marking, so it stops the nearest-producer walk despite not having caused the flow. Making it transparent recovers real causal edges—measured at 97 per episode on our single-component environment, with none lost—and is arguably the correct graph. It nonetheless degrades learning sharply (Section 6). We report this as a measured boundary of the specification rather than resolving it.*

4.2. Single-owner reward attribution

Each reward r_j is assigned to the *latest* decision on its lineage, $\text{own}(j) = \arg \max_{d \in \mathcal{A}(j)} u_d$, where $\mathcal{A}(j)$ is the set of decisions in j 's ancestry. Ownership is therefore a *partition*: every reward is counted exactly once. This is what makes the estimator a return decomposition rather than a hindsight Q -sample, in which each decision on a lineage receives the reward's full mass.

4.3. The recursion

Let $w(d \rightarrow s)$ be the share of s 's consumed tokens descending from d , so that token conservation gives

$$\sum_{d \in \text{pred}(s)} w(d \rightarrow s) = 1. \quad (1)$$

Define the row sum and the *successor-normalized* weights

$$R(d) = \sum_{s \in \text{succ}(d)} w(d \rightarrow s), \quad \hat{w}(d \rightarrow s) = \frac{w(d \rightarrow s)}{R(d)}, \quad \sum_{s \in \text{succ}(d)} \hat{w}(d \rightarrow s) = 1. \quad (2)$$

With $\rho(\Delta t) = e^{-\beta \Delta t}$ the SMDP discount applied at *causal* depth, the estimator is

$$A[d] = \text{owned}[d] - V[d] + \sum_{s \in \text{succ}(d)} \hat{w}(d \rightarrow s) \rho(u_s - u_d) (V[s] + \lambda A[s]), \quad (3)$$

evaluated by a single backward pass over $\mathcal{A}$ in decreasing clock order and emitted as $Q[d] = A[d] + V[d]$, so it is consumed exactly as every other credit scheme: the policy uses $A_d = Q[d] - V(s_d)$ and the value head regresses on $Q[d]$. Because V is fitted to the emitted Q , the critic converges to the same object the recursion bootstraps through and the fixed point is self-consistent. Algorithm 1 states the procedure.

4.4. Properties

- **Chain-exact.** Where every causal edge has out-degree one and unit weight, $\hat{w} \equiv 1$ and (3) is ordinary SMDP-GAE. Verified to floating point on real traces.
- **The mean variant is a special case.** Under equal inflow shares $\hat{w} = 1/|\text{succ}(d)|$, which is the mean-over-successors combination (CGAE). The mean is therefore not an underived rival but the equal-share instance of (3).
- **Cost.** Approximately one millisecond per episode; under 8% of a training iteration.
- **Postponement** receives the SMDP temporal-difference advantage $e^{-\beta \tau} V(s') - V(s)$ rather than a lineage credit, for consistent support.

Algorithm 1 Causal-order advantage estimation (CGAE-CF)

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1: input: decisions  $\mathcal{A}$  with clocks  $u_d$ ; token history; critic values  $V[\cdot]$ ;  $\lambda, \beta$ 
2: build  $\text{succ}(\cdot)$  and  $w(\cdot \rightarrow \cdot)$  by nearest-producer walk ▷  $O(|E|)$ 
3:  $\text{owned}[d] \leftarrow 0$  for all  $d$ 
4: for all rewards  $r_j$  do
5:    $\text{owned}[\text{own}(j)] += r_j$  ▷ latest decision on  $j$ 's lineage
6: end for
7: for all  $d \in \mathcal{A}$  in decreasing  $u_d$  do
8:   if  $\text{succ}(d) = \emptyset$  then
9:      $A[d] \leftarrow \text{owned}[d] - V[d]$ 
10:  else
11:     $R \leftarrow \sum_{s \in \text{succ}(d)} w(d \rightarrow s)$ 
12:     $A[d] \leftarrow \text{owned}[d] - V[d] + \sum_s \frac{w(d \rightarrow s)}{R} \rho(u_s - u_d)(V[s] + \lambda A[s])$ 
13:  end if
14: end for
15: return  $Q[d] = A[d] + V[d]$ 
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5. Theoretical analysis

5.1. A common form for the estimator family

Every variant we consider is a choice of edge coefficient $c(d \rightarrow s)$ in (3). Unrolling with $\lambda = 1$ and $V \equiv 0$,

$$Q[d] = \sum_{j: \text{own}(j) \in \text{desc}^*(d)} c_d(j) \rho(u_j - u_d) r_j, \quad c_d(j) = \sum_{\text{paths } d \rightsquigarrow \text{own}(j)} \prod c, \quad (4)$$

so a variant is characterized by the total path weight it places on each descendant's owned reward. Table 1 lists the family.

5.2. The causal partition, the estimand, and its unbiasedness

Everything below is stated relative to one reference estimator, which we define here and prove unbiased. Section 5.8 then measures how far the shipped estimator departs from it.

Descendants and causal components. Write $\text{desc}(d)$ for the set of decisions reachable from d under the successor relation of Section 4, and $\text{desc}^*(d) = \text{desc}(d) \cup \{d\}$ for its reflexive closure. Separately, build an *undirected* graph on $\mathcal{A}$ with an edge (d, d') whenever some reward r_j has both d and d' on its lineage, i.e. $\{d, d'\} \subseteq \mathcal{A}(j)$. Its connected components partition the decisions into *causal components* $C_1, \dots, C_M$; write $c(d)$ for the component of d . Because the whole ancestry of a reward is joined into one component, every reward with $\mathcal{A}(j) \neq \emptyset$ has a well-defined component $c(j) := c(d)$ for any $d \in \mathcal{A}(j)$; rewards with empty ancestry are policy-independent constants and drop out of the

Table 1: The estimator family, and the two sums that matter.

variant	$c(d \rightarrow s)$	$\sum_{\text{pred}(s)} c$	$\sum_{\text{succ}(d)} c$
CGAE-DAG	1 (closure, set semantics)	—	convex
CGAE-CF	w/R	unbounded	= 1 exactly
CGAE	$1/ \text{succ}(d) $	unbounded	= 1 exactly
CGAE-CAP	$w/\max(R, 1)$	≤ 1	≤ 1
CGAE-F	w	= 1 exactly	unbounded

gradient. Note $\text{desc}^*(d) \subseteq c(d)$ —a causal descendant shares a lineage with d , hence a component—so scoping credit to the descendant closure discards at least as much as scoping it to the component, and strictly more whenever a component holds two decisions neither of which descends from the other.

Definition 1 (The structural statistic K). *K is the number of causal components containing at least one reward, computed per episode and averaged over rollouts of an untrained policy. It is therefore a real number rather than an integer: on the N -copies benchmark $K = N$ holds per episode when every copy happens to emit a reward, and the reported means 2.0, 3.3 and 6.3 for $N = 2, 4, 8$ sit at or below N for that reason. No training and no reward model beyond the simulator’s own log is required to compute it.*

The two reward-to-go signals.. For decision d let

$$G_d^{\text{full}} = \sum_{j: t_j \geq u_d} \rho_d(t_j) r_j, \quad \Delta_d = \sum_{\substack{j: t_j \geq u_d \\ c(j) \neq c(d)}} \rho_d(t_j) r_j,$$

so that G_d^{full} is the ordinary SMDP return-to-go—the signal PPO and the lineage ablation mc-Q use—and Δ_d is the part of it generated by components other than d ’s. Write $\mathcal{F}_d$ for the history up to but excluding a_d , so that s_d is $\mathcal{F}_d$ -measurable and a_d is drawn given $\mathcal{F}_d$. We take as given the policy-gradient theorem in its causality (GPOMDP) form, that $\sum_d \nabla \log \pi(a_d | s_d)(G_d^{\text{full}} - V(s_d))$ is unbiased for $\nabla_{\theta} J$; every claim below is relative to it.

Assumption 1 (Component mean-exogeneity). *For every decision d , $\mathbb{E}[\Delta_d | \mathcal{F}_d, a_d] = \mathbb{E}[\Delta_d | \mathcal{F}_d]$: given the pre-action history, the expected cross-component future reward does not depend on which action is taken at d .*

This is the assumption referred to throughout as (A1), and A-E PN semantics argue for it rather than merely permitting it. A reward r_j is a deterministic function of the tokens consumed at its firing, and each of those tokens’ entire histories is recorded in the provenance DAG; hence r_j can be causally influenced by a_d only if some descendant of a_d lies on j ’s lineage—only if $d \in \mathcal{A}(j)$, which by construction of the partition means $c(j) = c(d)$. A cross-component reward therefore has no causal path from a_d , and neither its value nor its component membership can be altered by a_d (a lineage link to d would have placed it in $c(d)$ in the first place). The one residual channel is the exogenous randomness driving cross-component events—arrival types, service durations—and Assumption 1 asserts

that those draws are, in conditional mean, unaffected by a_d . Under the standard simulation model of independent per-firing draws this holds; it is a statement about the noise model, not about problem-specific structure. It can fail if a shared, finite resource pool couples nominally separate streams, which is precisely the $K = 1$ regime our negative control occupies.

The closure estimand.. The first row of Table 1 is the $c \equiv 1$ member of the family: the $\lambda = 1$, $V \equiv 0$ estimator that credits d with each descendant’s owned reward exactly once, with no path weighting at all,

$$A^*[d] = \sum_{j: \text{own}(j) \in \text{desc}^*(d)} \rho_d(t_j) r_j. \quad (5)$$

It is realized by CGAE-DAG *by construction* rather than to within a measured tolerance: at $\lambda = 1$ with $V \equiv 0$ its bootstrap term vanishes and what the recursion computes is (5) itself. We verify the implementation matches the definition—the two agree to 0 units in the last place over 191 decisions—but this is a check on code, not an estimate of a quantity. Because ownership is a partition (Section 4), each reward enters (5) at most once even when its owner is reachable from d along several distinct paths—the set semantics is what distinguishes it from the path-summing variants, and it is why a re-convergent diamond’s reward is counted once rather than twice.

Proposition 1 (The closure estimand is unbiased). *Under Assumption 1, the estimator $\hat{g}^* = \sum_d \nabla_\theta \log \pi_\theta(a_d | s_d)(A^*[d] - V(s_d))$ satisfies $\mathbb{E}[\hat{g}^*] = \nabla_\theta J(\theta)$ for any $\mathcal{F}_d$ -measurable baseline V .*

Proof. Write the discarded mass as $G_d^{\text{full}} - A^*[d] = \Delta_d + \Xi_d$, where Δ_d collects the cross-component rewards and $\Xi_d = \sum \rho_d(t_j) r_j$ runs over the rewards j with $t_j \geq u_d$, $c(j) = c(d)$ and $\text{own}(j) \notin \text{desc}^*(d)$: those in d ’s own component that the closure nonetheless omits. It suffices to show each vanishes in the gradient.

Fix d and condition on $\mathcal{F}_d$. Since a_d is drawn given $\mathcal{F}_d$ and both quantities are realized afterwards,

$$\mathbb{E}[\nabla \log \pi_\theta(a_d | s_d) X_d | \mathcal{F}_d] = \mathbb{E}_{a_d}[\nabla \log \pi_\theta(a_d | s_d) \mathbb{E}[X_d | \mathcal{F}_d, a_d]]$$

for $X_d \in \{\Delta_d, \Xi_d\}$. For $X_d = \Delta_d$ the inner expectation is free of a_d by Assumption 1. For $X_d = \Xi_d$, a reward j in that sum has $\text{own}(j) \notin \text{desc}^*(d)$; since $\text{own}(j)$ is the *latest* decision on j ’s lineage and the successor relation is time-forward, no decision of $\mathcal{A}(j)$ can lie in $\text{desc}^*(d)$ at all, so no descendant of a_d is on j ’s lineage. By the same A-E PN argument that justifies Assumption 1— r_j is a deterministic function of the tokens consumed at its firing, whose histories are recorded— r_j is then causally uninfluenced by a_d , and its conditional mean is likewise free of a_d .

In both cases the inner expectation is some $\bar{X}_d$ not depending on a_d , so the outer expectation is $\bar{X}_d \mathbb{E}_{a_d}[\nabla \log \pi_\theta(a_d | s_d) | \mathcal{F}_d] = \bar{X}_d \nabla_\theta \sum_a \pi_\theta(a | s_d) = \bar{X}_d \nabla_\theta 1 = 0$ by the score-function identity. The baseline term contributes zero by the same identity. Summing over d and taking total expectation gives the claim. $\square$

Remark 2 (What Proposition 1 does and does not license). *It licenses (5) as a reference point, and with it every relative statement in Sections 5.6 and 5.8. It says nothing about the estimator we ship, which is biased relative to (5) by a quantity Proposition 4 bounds; and it says nothing about performance—CGAE-DAG computes (5) exactly and is the second-worst arm we measured (Section 6). Unbiasedness here is a property of the estimand, not a recommendation.*

5.3. Why the bootstrap coefficient must be bounded

Proposition 2 (Successor normalization). *For the recursion (3) with coefficient c , the coefficient multiplying the bootstrap term at d is $\sum_{s \in \text{succ}(d)} c(d \rightarrow s)$. The predecessor constraint (1) does not bound this quantity. Writing $\hat{G}(d) = \sum_s \hat{w} \rho(V[s] + \lambda A[s])$ for the estimator’s own bootstrapped continuation value, the unnormalized and normalized forms differ by*

$$A_{\text{CGAE-F}}[d] - A_{\text{CGAE-CF}}[d] = (R(d) - 1) \hat{G}(d). \quad (6)$$

The proof is immediate from $w = R\hat{w}$. Three consequences drive everything in Section 6.

(i) *The error scales with the critic, not the reward.* $\hat{G}(d) = O(V) = O(H\bar{r})$ for remaining causal horizon H and mean per-decision reward $\bar{r}$, whereas the action-attributable signal is $\text{owned}[d] = O(\bar{r})$. Hence $|\text{error}|/|\text{signal}| \approx |R(d) - 1| \cdot O(H)$. Because the error grows with V , the estimator degrades as the critic improves—a divergence rather than a bias.

(ii) *It does not cancel as a baseline.* $R(d)$ depends on the realized DAG downstream of d , hence on the action at d . Forking every action at a decision state under common random numbers, R is identical across actions at only 18.8% (four independent copies) and 6.2% (single component) of fork points. An action-dependent term does not factor out through the score-function identity.

(iii) *Successor normalization removes it identically.* $\sum_s \hat{w} = 1$ by construction, so $R \equiv 1$ and (6) vanishes exactly—not in expectation. The mean variant is likewise protected.

Remark 3 (The predecessor sum is a choice of estimand, not a defect). *It is tempting to read (1) as a second conservation requirement and conclude that only $c = w / \max(R, 1)$ is admissible. That is wrong. $\sum_{\text{pred}(s)} c > 1$ means each of s ’s joint causes is credited with the whole continuation; since s consumes a token from every predecessor, each is necessary, and crediting each in full is precisely the but-for counterfactual that difference-reward methods use. Nothing in the policy-gradient argument constrains advantages across different decisions to sum to anything. Empirically the distinction barely moves total credit mass: $\sum_d Q[d]/G$ is 4.74 for CGAE-F and 4.86 for CGAE-CF on the single-component environment. Forcing the predecessor sum down (CGAE-CAP) costs performance, as Section 6 shows.*

5.4. Variance reduction, and the role of K

The estimand of (5) discards Δ_d without moving the expected gradient. What it buys is variance, and the size of the purchase is governed by K . The following isolates that mechanism in the simplest model that exhibits it; it is an idealization, not a description of any particular environment.

Proposition 3 (Component-scoped variance). *Fix a decision d and suppose the future reward-to-go splits as $G_d^{\text{full}} = \sum_{k=1}^K R_k$ with d in component 1, where R_k is component k ’s discounted reward-to-go and, conditionally on s_d , the R_k are mutually independent with variances σ_k^2 . Let the baseline be any s_d -measurable $V(s_d)$. Then*

$$\text{Var}(G_d^{\text{full}} - V(s_d) \mid s_d) = \sum_{k=1}^K \sigma_k^2, \quad \text{Var}(R_1 - V(s_d) \mid s_d) = \sigma_1^2.$$

If the components have comparable variance $\sigma_k^2 \approx \sigma^2$, component-scoped credit reduces the per-decision advantage variance by a factor $\approx K$ and the gradient-noise standard deviation by $\approx \sqrt{K}$.

Proof. Conditionally on s_d the baseline is a constant, so it does not affect either variance. By conditional independence $\text{Var}(\sum_k R_k \mid s_d) = \sum_k \text{Var}(R_k \mid s_d) = \sum_k \sigma_k^2$, which is the first equality; the second is immediate. Under $\sigma_k^2 \approx \sigma^2$ the ratio is $K\sigma^2/\sigma^2 = K$, and standard deviation scales as the square root. $\square$

Remark 4 (The gap between the model and the estimator). *Two idealizations separate Proposition 3 from what CGAE-CF actually computes. Conditional independence across components is stronger than Assumption 1, which constrains only conditional means: (A1) is what buys unbiasedness, independence is what buys the variance factor, and an environment can satisfy the first while only approximately satisfying the second. And CGAE-CF scopes to $\text{desc}^*(d)$, which is contained in $c(d)$ and usually strictly smaller; so it discards more than the model does—which is why the empirical gain is not a clean function of K (Section 6.6). The proposition should be read as identifying the mechanism and its governing quantity, not as predicting an effect size.*

The scoping is directly observable. With $S(d) = |\text{desc}(d)|/|\{\text{decisions after } d\}|$, balanced components give $S \approx 1/K$, and on the scaling benchmark with N copies (where the copies are independent by construction, so $K \leq N$, with equality whenever every copy emits a reward; see Definition 1) S measures 1.000, 0.472, 0.221, 0.114 for $N = 1, 2, 4, 8$ against $1/N = 1.000, 0.500, 0.250, 0.125$. At $N = 1$, $S = 1.000$ exactly: the estimator provably performs no scoping, and indeed every method ties there.

Remark 5 (S is not K). *The single-component environment has $S = 0.474$, essentially equal to the two-copy benchmark’s 0.472, yet gains nothing while the latter gains. Scoping away half the timeline buys variance reduction only if what is removed is independent of what is kept. S measures how much is excluded; K measures how much of it was independent.*

5.5. Causal credit reduces total credit mass

A natural objection to (4) is that crediting every causal ancestor must inflate credit relative to the return $G = \sum_j r_j$. It does, but far less than the baseline it is compared against: under plain return-to-go a reward at decision k enters the advantage of all k preceding decisions, giving $\sum_t G_t \approx (n/2)G$ for n decisions. Measured, $\sum_d Q[d]/G$ is 39.78, 21.23 and 18.40 for return-to-go on our three main environments (against $n/2$ of 34.5, 18 and 14.5) versus 4.34, 4.32 and 4.86 for CGAE-CF—a four- to ninefold *reduction*, tracking $1/S$ closely. The residual is the variance-reduction mechanism viewed from the other side. Absolute scale is in any case removed by per-batch advantage standardization.

5.6. A bias bound for CGAE-CF

Proposition 2 says the unnormalized form is unusable. It does not say the normalized form is correct. This subsection bounds how far CGAE-CF departs from the closure estimand A^* of (4), i.e. from the variant whose unbiasedness

under Assumption 1 is Proposition 1. The bound is therefore *relative*: it says how much bias is added on top of that estimator, not that the result is unbiased.

Assumption 2 (Time-forward causal edges). $u_s > u_d$ for every causal edge $d \rightarrow s$.

Assumption 3 (Bounded score). $\|\nabla \log \pi(a \mid s)\| \leq B$ for all (s, a) visited.

Assumption 2 makes the DAG acyclic and, because $\rho_{ds} = e^{-\beta(u_s - u_d)}$, makes the discount telescope along any path: $\rho_{ds_1} \rho_{s_1 s_2} \cdots \rho_{s_{k-1} e} = \rho_{de}$, independently of which path is taken. It is worth stating explicitly because the implementation clamps negative time differences to zero, which would silently break telescoping were an edge ever to point backwards in time.

Lemma 1 (Path weights are visit probabilities). *Under Assumption 2, $\hat{W} = [\hat{w}(d \rightarrow s)]$ is row-stochastic on non-sink nodes and induces a Markov chain on $\mathcal{A}$ with sinks absorbing. For this chain,*

$$c_d(e) = \Pr[\text{the walk started at } d \text{ visits } e] \in [0, 1].$$

Proof. Row-stochasticity is (2). By Assumption 2 the DAG is acyclic, so a walk visits any node at most once; consequently the events {the walk's initial segment is π }, indexed by the distinct paths $\pi : d \rightsquigarrow e$, are pairwise disjoint and their union is exactly the event that the walk visits e . Each has probability $\prod_{(x \rightarrow y) \in \pi} \hat{w}(x \rightarrow y)$, and summing gives $c_d(e)$. A probability lies in $[0, 1]$. $\square$

Define the *re-convergence deficiency* at d , in worst-case and reward-weighted form,

$$\delta_d = \max_{e \in \text{desc}(d)} (1 - c_d(e)), \quad \bar{\delta}_d = \frac{\sum_{e \in \text{desc}^*(d)} (1 - c_d(e)) |\text{owned}[e]|}{\sum_{e \in \text{desc}^*(d)} |\text{owned}[e]|},$$

and $\delta = \max_d \delta_d$. Both are computable from a handful of rollouts without training.

Proposition 4 (Bounded bias). *Under Assumptions 2 and 3, with $\lambda = 1$ and $V \equiv 0$:*

- (i) $c_d(e) \in [0, 1]$ for every d and every $e \in \text{desc}(d)$;
- (ii) $|A[d] - A^*[d]| \leq \bar{\delta}_d \sum_{e \in \text{desc}^*(d)} |\text{owned}[e]| \leq \delta_d \sum_{e \in \text{desc}^*(d)} |\text{owned}[e]|$;
- (iii) the policy-gradient bias relative to the A^* estimator obeys

$$\|\nabla J_{\text{CGAE-CF}} - \nabla J_\star\| \leq B \delta \sum_e |\text{owned}[e]| \cdot |\text{anc}^*(e)| \leq B \delta n G_{\text{abs}},$$

with $G_{\text{abs}} = \sum_j |r_j|$;

- (iv) if every node of $\text{desc}^*(d)$ has out-degree at most one, then $\delta_d = 0$ and $A[d] = A^*[d]$ exactly. More generally $\delta_d = 0$ if and only if $\text{desc}^*(d)$ is totally ordered by reachability.

Proof. (i) is Lemma 1. For (ii), telescoping of ρ gives $A^*[d] - A[d] = \sum_e (1 - c_d(e)) \rho_{de} \text{owned}[e]$; take absolute values, use $\rho \leq 1$ and $0 \leq 1 - c_d(e) \leq \delta_d$, and note the middle expression is exactly $\bar{\delta}_d$ times the total owned mass. For (iii), the

two estimators differ only through the summand, so $\nabla J_{\text{CGAE-CF}} - \nabla J_{\star} = \mathbb{E}[\sum_d (A[d] - A^*[d]) \nabla \log \pi_d]$; apply the triangle inequality, then (ii) and Assumption 3, and reindex the double sum by owner, each e being counted once per ancestor. For (iv), if out-degree is at most one throughout $\text{desc}^*(d)$ the walk is deterministic and visits every descendant, so $c_d(e) = 1$; conversely $c_d(e) = 1$ for all e says every maximal walk from d meets every descendant, which is total order by reachability. $\square$

The bound is instrumented, and its equality case is observed. Table 3 reports mean $c_d(e)$ per environment. On multi-site routing every decision has out-degree exactly one, and $c_d(e) = 1.0000$ with $\delta = 0$: CGAE-CF coincides with the unbiased closure estimand there, so the headline realistic result is produced by an unbiased estimator. Where out-degree exceeds one the deficiency appears and grows with fan-out: mean $1 - \bar{c}$ of 0.050 at four copies and 0.285 on the single-component environment. We verified Lemma 1 numerically by simulating the induced walk: Monte-Carlo visit frequencies match the closed form to ± 0.016 at 4000 samples, and $c_d(e)$ never left $[0, 1]$.

Remark 6 (What this does *not* say). *Proposition 4 bounds bias, not error, and certainly not performance. CGAE-DAG attains $\delta = 0$ by construction—it computes A^* exactly—and ranks second-worst of the nine arms in Table 6. The proposition explains why CGAE-CF is safe, in the precise sense that its departure from the unbiased estimand is bounded by a measurable structural quantity that vanishes on non-branching provenance graphs. It does not explain why it is good. Section 6 shows the two questions have different answers.*

5.7. What is still not proved

The bound above is stated for $\lambda = 1$ and $V \equiv 0$. With a critic and $\lambda < 1$ the recursion becomes, by the same telescoping argument,

$$A[d] = \mathbb{E}_{\text{walk}} \left[\sum_{k \geq 0} \lambda^k \rho_{d, X_k} \delta_{X_k} \right], \quad \delta_x = \text{owned}[x] - V[x] + \mathbb{E}_s [\rho_{xs} V[s]],$$

i.e. exactly GAE in expectation over the causal walk. The additional bias is then the ordinary GAE bootstrap bias from an imperfect critic, which is neither specific to this estimator nor bounded here.

5.8. A dimensionless relative bound

Proposition 4(iii) bounds an absolute quantity, and its bound scales with n because the gradient is a sum of n terms. That makes it weak. One cannot simply divide by $\|\nabla J_{\star}\|$: near a stationary point the denominator vanishes while the numerator need not, so no bound on the *ratio* of gradient norms can exist. The useful relative statement is about *direction*, and for this estimator it is available in closed form.

Assumption 4 (Nonnegative owned rewards). $\text{owned}[e] \geq 0$ for all e .

This holds in every environment we study—throughput and completion rewards are nonnegative, and we verified that no decision receives negative owned reward in any measured episode—but it is a property of the reward model,

not a theorem, and estimators for cost-minimization problems with mixed-sign rewards would need the general form of Proposition 4(ii) instead.

Define the *reward-weighted deficiency* and its multiplier

$$\bar{\delta}_d = \frac{\sum_e (1 - c_d(e)) \rho_{de} \text{owned}[e]}{\sum_e \rho_{de} \text{owned}[e]}, \quad m_d = 1 - \bar{\delta}_d \in [0, 1],$$

for every d with $A^\star[d] \neq 0$.

Proposition 5 (Exact rescaling and an exact cosine law). *Under Assumptions 2 and 4, with $\lambda = 1$, $V \equiv 0$:*

- (i) **Exact rescaling.** $A[d] = m_d A^\star[d]$ for every d .
- (ii) **The level of the shrinkage is free.** If $m_d \equiv m$ is constant, then $\nabla J_{\text{CGAE-CF}} = m \nabla J_\star$: the ascent direction and the set of stationary points are identical, and the factor m is absorbed by the step size—and removed outright by the per-batch advantage standardization used throughout. Only the dispersion of m_d can distort the direction.
- (iii) **Cosine law.** Let $p_d \propto A^\star[d]^2$ and let CV be the coefficient of variation of m under p . Then

$$\cos \angle(A, A^\star) = (1 + \text{CV}^2)^{-1/2} \quad \text{exactly,} \quad \text{hence} \quad 1 - \cos \angle \leq \frac{1}{2} \text{CV}^2.$$

- (iv) CV is dimensionless, independent of n , and computable from a handful of untrained rollouts.

Proof. (i) By Assumption 4 the denominator of $\bar{\delta}_d$ is $A^\star[d]$ and its numerator is $A^\star[d] - A[d]$, so $\bar{\delta}_d = 1 - A[d]/A^\star[d]$, which rearranges to (i). (ii) is immediate from linearity of ∇J in the per-decision coefficient. For (iii), write $u = A^\star$ and $A_d = m_d u_d$. Then $\langle A, u \rangle = \sum_d m_d u_d^2 = \mathbb{E}_p[m] \|u\|^2$ and $\|A\|^2 = \sum_d m_d^2 u_d^2 = \mathbb{E}_p[m^2] \|u\|^2$, so

$$\cos \angle = \frac{\mathbb{E}_p[m]}{\sqrt{\mathbb{E}_p[m^2]}} = \frac{\mathbb{E}_p[m]}{\sqrt{\mathbb{E}_p[m]^2 + \text{Var}_p(m)}} = (1 + \text{CV}^2)^{-1/2},$$

and $(1 + x)^{-1/2} \geq 1 - x/2$ gives the bound. $\square$

Measured.. Table 2 reports both sides. The identity (i) holds to machine precision (maximum absolute deviation 1.8×10^{-15} over all episodes), and the cosine law (iii) is exact to 6.7×10^{-16} —these are verifications of algebra, not fits. The practical content is the last column: the direction of the credit vector is preserved to within 11% in the worst case, and *exactly* on multi-site, where out-degree is one so $\bar{\delta} \equiv 0$.

Remark 7 (What the cosine law does and does not cover). *Proposition 5(iii) is a statement about the vectors of per-decision coefficients A and $A^\star$, not directly about the gradients $\sum_d A_d \nabla \log \pi_d$. The two coincide when the score vectors $\{\nabla \log \pi_d\}$ are orthonormal; in general they differ by a factor governed by the conditioning of the score Gram matrix, which we do not bound. We therefore read Table 2 as characterizing the estimator, not as a guarantee on the realized gradient. As with Proposition 4, everything is stated relative to $A^\star$, whose unbiasedness rests on Assumption 1.*

Table 2: Proposition 5 measured on untrained rollouts ($\lambda = 1$, $V \equiv 0$, $\beta = 0$). CV is the A^{*2} -weighted coefficient of variation of the multiplier.

environment	$\bar{\delta}$ (mean)	CV	$\cos \angle$ measured	$(1 + CV^2)^{-1/2}$	$1 - \cos \angle$
multi-site	0.000	0.000	1.00000	1.00000	0.000
$N=4$	0.095	0.195	0.97823	0.97823	0.022
$N=8$	0.109	0.500	0.89106	0.89106	0.109
single comp.	0.374	0.423	0.91987	0.91987	0.080

6. Experiments

6.1. Protocol, and a budget caveat that changes what we report

All comparisons use common random numbers: every greedy evaluation point, for every seed and method, scores on one fixed scenario set. On the single-component environment the measured evaluation noise floor is ± 0.231 standard deviations per 20-episode point and ± 0.327 on a paired difference. Returns are normalized as $(\text{score} - \text{random}) / (\text{heuristic} - \text{random})$. We report paired t -tests and Wilcoxon signed-rank tests, and count *collapses* (normalized final ≤ 0.25). Every arm uses identical hyperparameters, with no per-arm tuning; Appendix Appendix C gives them, the architecture, and the evaluation and seeding protocol in full, and Appendix Appendix A defines every comparator arm.

Training budget.. An earlier protocol trained the scaling benchmark for 15 epochs. At that budget the policy has barely moved—final entropy 0.91–0.94 from 1.00, with only 19–38% of the random-to-heuristic headroom captured—and the comparison is not converged. It produced a significant and entirely spurious result: at two copies, CGAE-CF measured 0.401 against CGAE-F’s 0.837 (4W/14L, $p = 0.0043$), which at 40 epochs becomes +0.102 ($p = 0.51$), sign reversed. **We therefore report the 40-epoch budget throughout and retain the 15-epoch cells only as a budget-sensitivity ablation.** Three independent measurements confirm the artifact: a λ sweep (0.401 $\rightarrow$ 0.579 $\rightarrow$ 0.639 as λ falls 0.95 $\rightarrow$ 0.85 $\rightarrow$ 0.70 toward CGAE-F’s effective 0.851), near-identical credit vectors (mean $|Q|$ 2.22 vs 2.23), and bimodal outcomes with nothing in between.

6.2. Environments and their structure

We use three environment families: a multi-site skills-based routing problem with dedicated per-site specialists, which is the realistic case; an N -copies scaling benchmark of N structurally independent assignment problems sharing one policy, in which the independence structure is a controlled parameter; and a two-stage service line whose single shared employee pool links every case into one causal component, which is the negative control. Appendix Appendix B specifies all three, together with the random and heuristic anchors of the normalization.

Figure 3 shows the learning curves at the two ends of this range, and Tables 4 and 5 give the final numbers. Two features of Table 3 are used repeatedly. First, K predicts whether scoping can help. Second, fan-out predicts whether

Table 3: Structural statistics, computed from random rollouts before training. K = reward-bearing causal components; fan-out = $\text{mean } |\text{succ}(d)|$; S = scoping ratio.

environment	K	top comp. share	fan-out	S	character
multi-site routing	8.0	15.5%	1.000	0.106	realistic, most decomposable
$N = 8$ copies	6.3	35.5%	1.160	0.114	scaling endpoint
$N = 4$ copies	3.3	42.4%	1.137	0.221	scaling benchmark
$N = 2$ copies	2.0	56.1%	1.020	0.472	low- K point
single component	1.0	100%	1.387	0.474	negative control

the successor-combination rule matters: on the multi-site environment fan-out is exactly 1.000 across all measured decisions, so CGAE-CF, CGAE and CGAE-F are **bit-identical** there (maximum absolute credit difference 0.0), a fact we verify rather than assume.

6.3. Main results at the converged budget

Against PPO the effect is large where K is large.. On multi-site, CGAE-CF improves normalized return by +0.512 (18W/1L, $p < 0.0001$) and collapses on 0 of 20 seeds against PPO’s 14. At eight copies the paired gain is +0.194 ($p = 0.0019$), at four copies +0.264 ($p = 0.0009$), and against CFGAE at four copies +0.370 ($p = 0.0005$). At two copies the gain is +0.080 and not significant ($p = 0.34$), and at $K=1$ it is -0.030 ; the effect is therefore present at the three high- K configurations and absent at the two low- K ones, which is the pattern Proposition 3 predicts and which we return to in Section 6.6.

The gain is located in the causal restriction, not the discounting.. The lineage ablation mc-q is the identical SMDP return-to-go with the causal restriction removed and everything else—discounting, normalization, optimizer—held fixed. It loses to CGAE-CF on every environment where scoping is possible, and by margins that track K : +0.642 on multi-site (20W/0L, $p < 0.0001$, where mc-q scores 0.030 and collapses on 17 of 20 seeds), +0.418 at four copies (18W/2L, $p < 0.0001$) and +0.230 at two copies (10W/4L, $p = 0.010$). On the single-component environment, where Proposition 3 says there is nothing to scope, the same ablation is null: +0.024, $p = 0.52$. The ablation therefore separates exactly where the mechanism says it should and nowhere else.

Against other bounded causal schemes the effect is null everywhere.. At four copies CGAE-CF versus CGAE, CCF, CGAE-CAP and CGAE-F spans -0.015 to $+0.009$, every $p \geq 0.43$; on the single-component environment -0.040 to $+0.056$, every $p \geq 0.27$; at two copies $+0.080$ to $+0.204$, every $p \geq 0.052$. (The comparison set here is the bounded causal schemes; the non-causal ablation mc-q is significant at two and four copies, as reported above.) We state this explicitly because an earlier version of this work, run at a shorter budget with fewer seeds, reported a significant advantage over CCF; that difference was a convergence-speed artifact—ccf gains +0.24 at four copies under the longer budget—and it

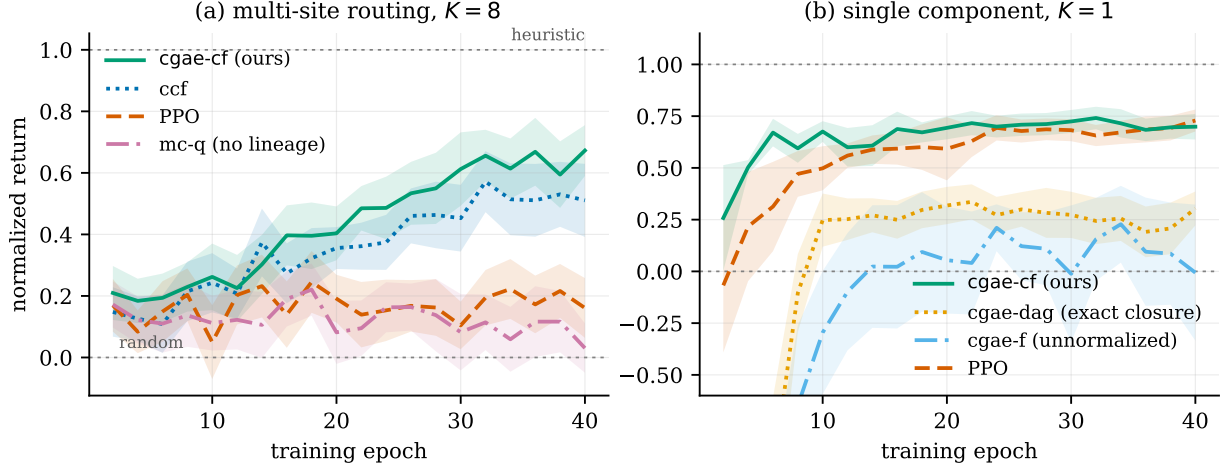


Figure 3: Learning curves, mean over 20 common-random-number seeds with 95% confidence bands; every series carries its own dash pattern so the figure reads in greyscale. **(a)** Where the problem decomposes ($K=8$), causal-order credit separates from PPO early and holds the gap; the lineage ablation mc-q—the same return-to-go without the causal restriction—tracks PPO, locating the gain in the decomposition rather than in the discounting. **(b)** Where it does not ($K=1$), CGAE-CF and PPO are indistinguishable, which is the intended negative control. The two failing variants separate sharply here: CGAE-DAG plateaus at a third of the attainable return, and CGAE-F climbs, turns over, and ends near random—the divergence Proposition 2 predicts for an unbounded bootstrap coefficient. Its early confidence band extends below the axis.

does not survive. Multi-site is the one borderline case ($+0.162$, $p = 0.0500$, 11W/9L), which we read as no reliable difference given the disagreeing rank test.

On the negative control the method is null, as designed. Against PPO on the single-component environment, CGAE-CF is -0.030 ($p = 0.46$), and indeed *seven* arms lie within 0.06 of one another there. The mechanism predicts this: $K = 1.0$ and one component holds 100% of reward mass, so there is nothing to scope. A method that predicts its own failure case is the strongest evidence we can offer for the mechanism, and the value of the environment is that it is where the *ablations* separate (Section 6, Table 5).

The unbounded coefficient fails on the highest- K environment too. At the shorter budget and five seeds, CGAE-F appeared healthy at eight copies (0.994 ± 0.014). At 20 seeds it measures 0.741 ± 0.654 with 3 collapsed seeds, against CGAE-CF’s 0.999 ± 0.025 and no collapses ($+0.258$, $p = 0.0999$). Together with the single-component result ($+0.705$, $p = 0.0006$), this establishes that the divergence of Proposition 2 is not a property of one environment: it appears at both ends of the decomposability range, and it is invisible at small seed counts because it manifests as occasional catastrophic runs rather than as a uniformly lower mean.

6.4. The successor normalization is necessary

Two rungs of the ladder are significant and the rest are not. The predecessor-normalized form CGAE-F and the exact-closure form CGAE-DAG both lose decisively; the choice between flow-weighted, uniform, and sub-stochastic

Table 4: Normalized final return at the converged 40-epoch budget, 20 common-random-number seeds for every cell. Parenthesised values are standard deviations; *coll.* counts collapsed seeds (normalized ≤ 0.25) out of 20. Dashes are cells not run. Bold marks the best arm in a row; on the single-component environment no arm is significantly best and none is bolded. On multi-site the causal fan-out is exactly 1, so CGAE-CF, CGAE and CGAE-F are provably the same algorithm and their three columns are bit-identical—the CGAE-F entry there is that shared value, not a separate run.

environment	K	CGAE-CF	CGAE	CCF	CGAE-CAP	CGAE-F	MC-Q	PPO
multi-site	8.0	0.673	0.673	0.511	—	0.673	0.030	0.161
		(0.19)	(0.19)	(0.27)		(0.19)	(0.18)	(0.22)
	coll.	0	0	3	—	0	17	14
$N=8$	6.3	0.999	—	0.977	—	0.741	—	0.804
		(0.03)		(0.06)		(0.65)		(0.23)
	coll.	0	—	0	—	3	—	0
$N=4$	3.3	0.980	0.995	0.974	0.971	0.989	0.562	0.715
		(0.09)	(0.03)	(0.10)	(0.10)	(0.03)	(0.29)	(0.27)
	coll.	0	0	0	0	0	5	1
$N=2$	2.0	0.882	0.797	0.799	0.761	0.751	0.651	0.802
		(0.18)	(0.30)	(0.30)	(0.43)	(0.34)	(0.39)	(0.30)
	coll.	1	3	3	3	4	5	3
single comp.	1.0	0.699	0.643	0.727	0.722	−0.006	0.675	0.729
		(0.14)	(0.19)	(0.13)	(0.15)	(0.75)	(0.10)	(0.12)
	coll.	0	1	0	0	10	0	0

predecessor handling does not matter.

CGAE-F diverges with training, exactly as Proposition 2 predicts.. The signature is visible within a single budget, without appeal to a second one. Averaged over the same 20 seeds, CGAE-F reaches a best-checkpoint value of 0.443 and ends at −0.006: a within-run drop of 0.449, against 0.089–0.136 for every arm that learns and 0.212 for CGAE-DAG. It is not that CGAE-F fails to find a good policy—it finds one and then destroys it. Ten of its 20 seeds end collapsed while only two ever collapse at their best checkpoint. One seed’s greedy curve rises to 10.3, holds for six evaluation points, then falls to 0.15—normalized −1.97, far below random—while policy entropy stays flat at 0.18: a confident collapse, not an exploration failure. Extending the budget makes it worse rather than better, which is what an error term proportional to the critic must produce: the estimator worsens as the critic improves.

Reporting metric.. Tables 4 and 6 report final greedy return, as everywhere else in this paper. Because this environment is documented as drift-prone, and because final return is the metric most contaminated by scenario noise—which is why best-checkpoint restoration exists in the first place—we also give the best-checkpoint reading, under which both significant results and the whole ordering are unchanged: CGAE-F 0.443_(0.14) (+0.390 against CGAE-CF, 20W/0L, $p < 0.0001$) and CGAE-DAG 0.516_(0.15) (+0.316, 19W/0L, $p < 0.0001$), against CGAE-CF’s 0.833_(0.08), with every remaining

Table 5: Paired comparisons against CGAE-CF (20 seeds, paired t -test). The pattern is uniform: large and significant against non-causal or unbounded-coefficient baselines, null against every bounded causal variant.

environment	comparator	difference		p	
multi-site	PPO	+0.512	< 0.0001	***	
multi-site	MC-Q	+0.642	< 0.0001	***	
multi-site	CCF	+0.162	0.0500	borderline	
$N=8$	PPO	+0.194	0.0019	**	
$N=8$	CGAE-F	+0.258	0.0999		
$N=8$	CCF	+0.021	0.24	null	
$N=4$	MC-Q	+0.418	< 0.0001	***	
$N=4$	CFGAE	+0.370	0.0005	***	
$N=4$	PPO	+0.264	0.0009	***	
$N=4$	CGAE, CCF, CGAE-CAP, CGAE-F	−0.015 to +0.009	≥ 0.43	null	
$N=2$	MC-Q	+0.230	0.0100	*	
$N=2$	all other arms	+0.080 to +0.204	≥ 0.052	null	
single comp.	CGAE-F	+0.705	0.0006	***	
single comp.	CGAE-DAG	+0.395	< 0.0001	***	
single comp.	PPO, CGAE, CGAE-CAP, CFGAE, CCF, MC-Q	−0.040 to +0.056	≥ 0.27	null	

arm within 0.041 of CGAE-CF and no p below 0.13. The magnitudes shrink under best-checkpoint precisely because checkpoint restoration is a partial guard against the late-training divergence described above; no conclusion in this section depends on the choice.

6.5. Computing the estimand exactly does not pay

CGAE-DAG computes (5) by construction: it reproduces textbook GAE on a chain to machine precision, and it counts a re-convergent diamond’s reward once rather than twice. It is the unbiased member of the family. It ranks second-worst of nine.

The structural reason is visible in Table 3. On a single-component environment the descendant closure is most of the remaining episode ($S = 0.474$, and 0.712 on a related environment), so the exact estimand approaches the unscoped return and exactness buys no scoping at all. Exactness is a property of the target, and on a one-blob environment that target barely varies with the action.

The weighted variants under-recover the estimand, and the shortfall compounds with causal depth.. Every path-weighted variant places total weight $c_d(j) \leq 1$ on a descendant’s owned reward, so it recovers only a fraction of (5). Measured at $\lambda = 1$, $V \equiv 0$, $\beta = 0$ over five random rollouts, the mean recovered fraction is 0.58–0.63 on the single-component environment and 0.89–0.91 at four copies, against CGAE-DAG’s 1 by construction, and it decays

Table 6: Single-component environment ($K=1$), 20 seeds, normalized final return, paired against CGAE-CF. Only two rungs of the ladder are significant; every bounded variant—PPO and the lineage ablation MC-Q included—is indistinguishable from every other.

variant	final (SD)	collapses	paired vs CGAE-CF
CFGAE	0.739 _(0.11)	0	−0.040 ($p = 0.27$)
PPO	0.729 _(0.12)	0	−0.030 ($p = 0.46$)
CCF	0.727 _(0.13)	0	−0.028 ($p = 0.54$)
CGAE-CAP	0.722 _(0.15)	0	−0.023 ($p = 0.55$)
CGAE-CF	0.699 _(0.14)	0	—
MC-Q	0.675 _(0.10)	0	+0.024 ($p = 0.52$)
CGAE	0.643 _(0.19)	1	+0.056 ($p = 0.36$)
CGAE-DAG	0.304 _(0.19)	8	+0.395 ($p < 0.0001$)
CGAE-F	−0.006 _(0.75)	10	+0.705 ($p = 0.0006$)

monotonically with causal depth—for CGAE-CF on the single-component environment, from 1.00 at depth 1 to 0.49 at depth 9.

We report this as a property of the estimators rather than as an explanation of the ordering, because on its own it cannot be one. Advantages are standardized per batch before each policy update (Appendix Appendix C), so a *uniform* shrinkage of all credit is invisible to the gradient by construction; only the dispersion of c_d can act, and that dispersion does not order the variants either—on the single-component environment CGAE-F has the *lowest* coefficient of variation of the four (0.42 against 0.49–0.64) and is the arm that collapses. What the measurement does establish is the fan-in dilution that motivates CGAE-DAG in the first place, and that the dilution is systematic in depth rather than incidental.

What we could not explain.. The ordering of the four working arms is not accounted for by any mechanism we tested. Six candidates were probed and all six failed, two of them *inverted*—per-state ranking fidelity against a common-random-number counterfactual put CGAE-DAG first ($\rho = +0.681$, top-1 agreement 83.3%) against CGAE-CF’s +0.435 and 33.3%, and credit signal-to-noise ratio likewise favoured the losing arm. Appendix Appendix D reports each probe with the measurement that rejected it. The only variable that separates working from failing arms without overlap is final policy entropy (0.27–0.56 against 0.13–0.15), and whether that is cause or symptom is untested; the decisive experiment is an entropy-bonus intervention, which we have not run. We also note that all six probes were static and used a random policy, which may be the wrong regime for the question. We therefore state the ordering as an open question. What the ladder does support—and all it needs to support—is the negative claim in Section 6.4: exactness is not what separates the working estimators from the failing one, and boundedness is.

6.6. Scaling

The gain over PPO is present at every $K > 1$ and absent at $K = 1$, but it is *not* monotone in K :

environment	K	gain over PPO	p
single component	1.0	−0.030	0.46
$N=2$ copies	2.0	+0.080	0.34
$N=4$ copies	3.3	+0.264	0.0009
$N=8$ copies	6.3	+0.194	0.0019
multi-site	8.0	+0.512	< 0.0001

We report the non-monotonicity rather than smoothing it, because its cause is visible and is not about K . At $N=8$ every causal arm saturates near the heuristic (0.977–0.999, standard deviations 0.03–0.06) and PPO itself reaches 0.805, leaving under 0.20 of headroom for any method to claim; the gain is compressed by the ceiling, not by weaker structure. The same effect, more mildly, explains $N=2$. **The defensible claim is therefore that K predicts *whether* component-scoped credit can help, not *how much*:** it is a screening statistic, and the magnitude additionally requires headroom. We do not fit a trend through these five points.

The restricted role of the $N=2$ configuration.. We use it as the low- K end of this curve and for nothing else. It cannot discriminate between estimators: at the converged budget the spread across all seven arms there is 0.169 with nothing significant, and at the shorter budget the same configuration produced a spread of 0.576 that *was* significant, in the wrong direction—the artifact that motivated the budget audit at the head of this section. A configuration that reverses its verdict between budgets and separates nothing at the longer one has too little headroom and too much seed variance to rank methods. It earns its place here only as a point on the K axis, and we draw no conclusion from the ordering of arms within it.

7. The boundary-condition investigation

Proposition 3 makes a sharp prediction: cGAE-CF’s variance reduction scales with the number of causally independent components K , and collapses to *exactly* PPO when $K = 1$ (a single shared resource pool links every case into one component). This is the theorem’s own honest null, not a failure. We use a two-stage service-line environment with a single shared, skill-heterogeneous employee pool as the decisive $K=1$ case. This is the same environment called *single component* in Section 6; here we call it the *bottleneck environment* to name the mechanism under test rather than the statistic. Its experiments predate the converged-budget re-runs and use a 30-epoch, 10-seed protocol, which is why the baseline arm reads 0.739 here against 0.729 in Table 6; see Appendix Appendix E, which also gives the technical detail of all three mechanisms. We ask: can the residual gap on this genuinely single-component problem be closed by any other provenance- or structure-based route? We tested three independent, well-motivated candidates—never

touching credit, only what the learner observes—and found each provably inert or empirically neutral. Table 7 and Figure 4 summarize, and Figure 5 gives the reward-shaping coefficient dial in full; we report all three not as a coda but as a result, since two of the three failures follow from a proof rather than a correlation and together they localize the bottleneck’s difficulty precisely.

Reward shaping. Potential-based shaping from PN topology (Ng et al., 1999), SMDP-generalized, is provably policy-invariant for any potential function—verified directly in code via an exact terminal-value cancellation test, not merely cited. A full coefficient dial on the bottleneck environment (same cached PPO baseline reused across all four points) shows reach-rate cleanly monotonic in the injected-variance direction ($8/10 \rightarrow 4/10 \rightarrow 1/10$ seeds reaching 95% of the heuristic anchor as the coefficient rises $0.2 \rightarrow 0.5 \rightarrow 1.0$), and no coefficient produces a significant win: 0.739 ± 0.10 (baseline) vs. 0.766 ± 0.10 ($p = .48$, coefficient 0.2) vs. 0.420 ± 0.10 , 0W/10L (coefficient 1.0). The mechanism is that constant exogenous arrivals make the potential rise as well as fall, injecting a per-step signal on the same order of magnitude as the full episode return—unbiased in aggregate (guaranteed by the theorem) but a substantially noisier value-regression target at a finite budget.

Structural input features. Static, conflict-graph-derived scalars appended to each action node’s own input features (does this action type structurally compete for a token, and with how many; and, after a first null result, the minimum topological distance to the nearest reward among an action’s required inputs) are both statistically neutral ($p = .58$ and $p = .69$ respectively). The stronger finding is *why*: every action node already carries a one-hot action-type identity at zero hops, so the two competing action types in the bottleneck environment are already fully distinguishable with or without any structural feature. Any scalar computed as a function of action type alone is therefore a deterministic function of information a single learned weight on the one-hot could already reconstruct—provably redundant, not merely observed to be unhelpful across two experiments, and true of this architecture on any topology, not only the bottleneck environment.

Actor architecture. We identify and prove a previously undocumented limitation: the actor decodes each action’s logit from only that action node’s own final message-passing embedding, unlike the critic, which has always pooled over all action/postpone node embeddings for its value estimate. Combined with graph edges directed strictly along token flow (no reverse edges at any real observation call site), an action’s logit can depend on state only via a forward path reaching it within the network’s depth. We prove this is a hard architectural fact, not a training-budget limitation: on a purpose-built topology of two structurally independent two-stage pipelines, perturbing one pipeline’s downstream marking left the other pipeline’s action logit *exactly, bit-for-bit unchanged* under a randomly initialized, untrained network ($\Delta = 0.000$)—reachability is a property of the computational graph, independent of learned weights. We fix this (concatenating the same pooled context the critic already computes onto each action node before decoding), confirm the fix closes the gap on the same proof topology ($\Delta = 0.000 \rightarrow 0.528$), and re-test on the bottleneck environment: norm. final 0.739 ± 0.10 (baseline) vs. 0.746 ± 0.14 (fixed architecture), $p = .91$ —no significant effect.

Table 7: The boundary-condition investigation, bottleneck environment ($K=1$).

mechanism	theoretical status	result
Reward shaping	policy-invariant, any potential	neutral (0.2) to harmful (1.0), 0W/10L
Structural input features	mechanically correct	neutral, $p = .58 \rightarrow .69$; provably redundant
Actor global context	proven to close a real gap	neutral, $p = .91$; gap was already marginal here

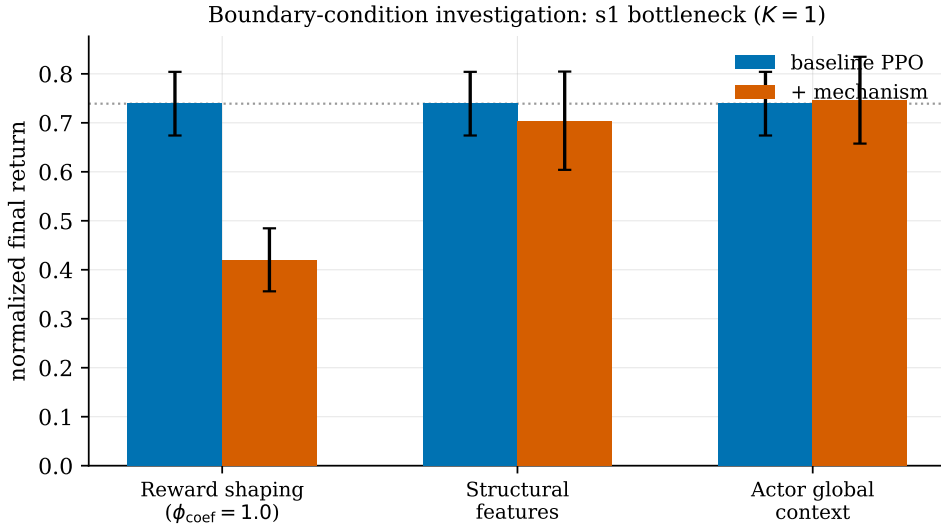


Figure 4: Baseline PPO vs. each of the three boundary-condition mechanisms on the bottleneck environment ($K=1$; error bars are 95% CIs over seeds). None significantly improves on the baseline; reward shaping at $\phi_{\text{coef}}=1.0$ is significantly worse.

The proof and the training result are not in tension: the bottleneck environment’s shared resource pool happens to recycle, giving the unfixed architecture a marginal, accidental backdoor to the same information, so the blind spot was never total there, and whatever fraction the fix newly exposes is not, in practice, decision-relevant enough to move a well-tuned baseline at this training budget.

None of this is a failure of experimental design: each mechanism does exactly what it was built to do. It is evidence about where the bottleneck environment’s difficulty does *not* live—not in what the learner can observe, and not, per Proposition 3, in an avoidable credit bias on this genuinely single-component topology. We state this as an open problem rather than a solved one in Section 8.

8. Discussion

A decision rule for practitioners.. Compute K from a handful of random rollouts before adopting anything. If the problem decomposes, even partially, into weakly coupled sub-streams—parallel lines, multiple sites, segmented resource pools—causal-order credit assignment gives real, cheap, drop-in gains with no hand-specified structure and

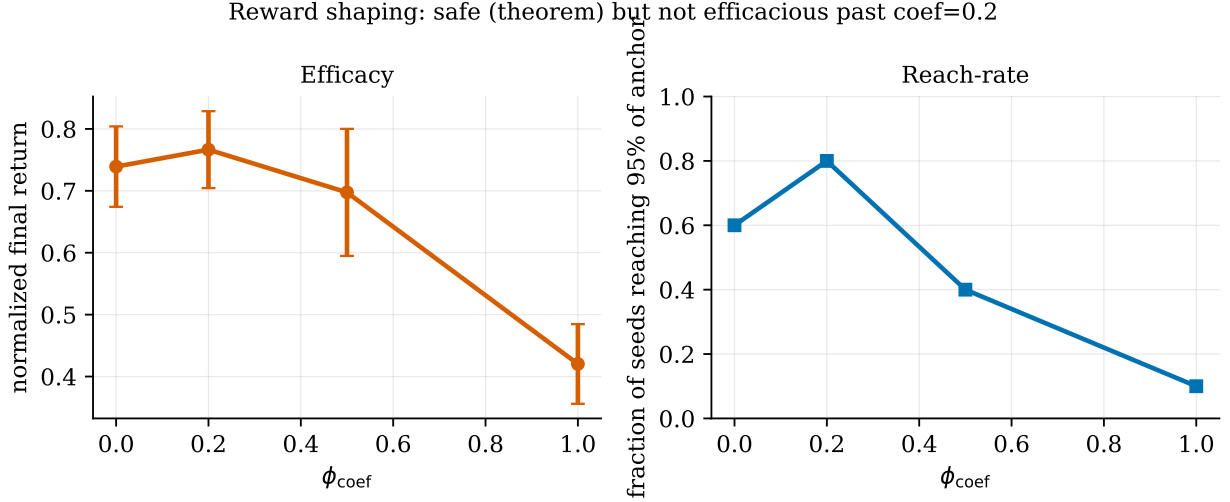


Figure 5: The reward-shaping coefficient dial on the bottleneck environment: efficacy (left) degrades once past $\phi_{\text{coef}} = 0.2$, and the fraction of seeds reaching 95% of the heuristic anchor (right) degrades monotonically—safety (Section 2, Ng et al., 1999) and efficacy are independent properties, and this mechanism has the first without the second at practical training budgets.

no forked simulation. If $K \approx 1$, expect a null; the method is *safe* there rather than harmful, which is what Table 6 establishes. Compute mean fan-out as well: where it is 1, every variant in Table 1 is the same algorithm and the implementation choice is irrelevant.

What the ablation ladder teaches.. Exactness is not the design objective; bounded error is. The precedent is GAE itself, whose $\lambda < 1$ settings are biased and universally preferred over the unbiased $\lambda = 1$ case. Our unbiased member performs worse than three biased ones, and the one variant that fails catastrophically is the one whose error is *unbounded*. We put this negatively on purpose: it says what does *not* determine the outcome. Which of the bounded variants to prefer is not something our evidence settles, and Section 6.5 records the six mechanisms we tested and rejected rather than offering a seventh we cannot support.

Limitations.. Five problem configurations drawn from three environment families. CGAE-CF is biased, and we verified the action dependence of R rather than assuming it away. The mechanism behind the ladder ordering is unexplained after the six falsified hypotheses of Appendix Appendix D. The bias bound of Section 5 is stated *relative* to the closure estimand A^* , whose own unbiasedness rests on Assumption 1; it is not an absolute guarantee. That assumption is argued from A-E PN semantics rather than verified per environment, and it is weakest exactly where a shared resource pool couples nominally separate streams. And we do not claim superiority over prior component-factored schemes at converged budgets.

What the method requires, and what it does not.. The scope boundary is not the problem domain but the reward structure. Nothing in the estimator is specific to task assignment: it needs only that rewards attach to discrete, token-

consuming firings whose provenance the simulator records, and that enough of the reward mass sit in distinct causal components for K to exceed one. Admission control and network revenue management satisfy the first condition directly, since revenue is earned at an acceptance, which is a firing that consumes a request token. Multi-item inventory control does not satisfy it straightforwardly: holding cost accrues per period rather than at a firing, so there is no consuming transition for the reward to attach to, and the cost would have to be discretized onto firings before ownership is even well defined. We have not run a second domain, so this states what the construction requires rather than where it has been shown to work.

9. Conclusion

An A-E PN simulation records token provenance for free, and that provenance turns hand-specified factored credit into a derived quantity. Running generalized advantage estimation along the provenance DAG rather than the trajectory index yields a cheap, fork-free policy gradient that improves substantially over discount-matched PPO wherever the problem’s causal structure is even partially decomposable, and is provably safe where it is not. The one design choice that matters is keeping the bootstrap coefficient bounded: the natural formulation does not, and diverges as a result.

Appendix A. Estimator definitions and additional remarks

Every arm compared in Section 6 differs from every other only in how the per-decision target $Q[d]$ is formed. All of them are consumed identically—the policy uses $A_d = Q[d] - V(s_d)$ and the value head regresses on $Q[d]$ —so the comparison isolates the credit rule and nothing else. Throughout, u_d is decision d ’s clock, t_j reward j ’s firing time, $\rho_d(t) = e^{-\beta \max(0, t - u_d)}$ the SMDP discount, $\mathcal{A}(j)$ the decisions on j ’s lineage, $c(\cdot)$ the causal-component map and $\text{own}(\cdot)$, $\text{succ}(\cdot)$, $\text{desc}^*(\cdot)$ as in Sections 4 and 5.2.

Non-causal reference points..

- **PPO** is the unmodified baseline: ordinary GAE along the trajectory index with the SMDP discount, $A_t = \delta_t + \gamma \lambda A_{t+1}$.
- **mc-q** (*the lineage ablation*) is the plain Monte-Carlo SMDP return-to-go, $Q[d] = \sum_{j: t_j \geq u_d} \rho_d(t_j) r_j$. Every decision sees every subsequent reward whether or not it caused it. It shares the estimator *form* of the causal Monte-Carlo schemes ($\lambda = 1$, no bootstrap, wall-clock discount) and differs from them only by dropping the causal restriction, which is exactly what makes it the ablation that prices the lineage. It needs no token DAG.

Component-scoped schemes.. These restrict *which* rewards a decision sees, keeping the Monte-Carlo return-to-go.

- LRQ credits d with the full discounted mass of every reward on whose lineage it sits, $Q[d] = \sum_{j: d \in \mathcal{A}(j)} \rho_d(t_j) r_j$. A reward with k lineage decisions contributes its whole mass to all k of them, so this is a per-decision hindsight Q -sample and *not* a return decomposition; it must never be summed as a return. It is the scheme single-owner attribution (Section 4) replaces.
- ccf builds the realized causal-component partition of Section 5.2 by union-find over each reward's lineage, then credits $Q[d] = \sum_{j: c(j)=c(d), t_j \geq u_d} \rho_d(t_j) r_j$: its own component's return-to-go. Because the partition is computed from the *realized* episode, component membership can itself depend on the action taken.
- s-ccf replaces that realized partition with a static, topology-only one derived from the net's structure, which is action-invariant by construction. It is reported here for completeness; it does not appear in the main tables.

Component-filtered GAE. cFGAE takes ccf's partition but consumes it through PPO's own recursion rather than as a Monte-Carlo sample: for each component it runs the ordinary SMDP-GAE over the step-reward stream masked to that component, and each decision reads the advantage from its own component's pass. Its $K = 1$ limit is therefore PPO *exactly*—the mask is identically one and the recursion is term-for-term PPO's—whereas every Monte-Carlo scheme above degenerates at $K = 1$ to mc-q, a different estimator that merely happens to score similarly. That exactness is why cFGAE is the right comparator for "does the *partition* help, holding the estimator form fixed".

The causal-order family. These run GAE's recursion along the provenance DAG, differing only in the edge coefficient $c(d \rightarrow s)$ of (3); Table 1 collects the coefficients and the two sums that matter.

- CGAE-CF (*the method*) uses $\hat{w} = w/R$, normalized over successors.
- CGAE uses $1/|\text{succ}(d)|$, the equal-share special case.
- CGAE-F uses the raw flow weight w , normalized over predecessors—the natural formulation, and the one Proposition 2 shows to be unbounded.
- CGAE-CAP uses $w/\max(R, 1)$, which only ever divides, so both the predecessor and successor sums are at most one. It prices the share-versus-but-for choice of Remark 3.
- CGAE-DAG abandons path weighting and uses set semantics: it credits each decision with each descendant's owned reward exactly once, computing A^* of (5) directly. It is the unbiased member of the family (Proposition 1).
- CGAE-CF2 is not an ablation of any design choice but a different DAG construction: CGAE-CF with POSTPONE made transparent to the nearest-producer walk. It appears only in the discussion of that specification wrinkle.

Postponement. In every causal scheme a POSTPONE decision receives, in place of a lineage credit, the SMDP temporal-difference advantage $e^{-\beta\tau}V(s') - V(s)$, so that every arm gives postponement consistent support.

Remark 8 (Why single-owner attribution rather than LRQ’s lineage sample). LRQ gives each reward’s full mass to every decision on its lineage, so summing its output over decisions overcounts the return by a factor that grows with lineage depth. Single-owner attribution makes ownership a partition, which is what turns the estimator into a return decomposition and lets the unrolled form (4) be read as a weighting of each reward. It also removes a bias LRQ carries in the opposite direction: restricting credit to one lineage drops the rewards that sibling lineages—the roads not taken at d —would have earned, and those are causal descendants of alternative actions at d , hence not mean-independent of a_d .

Remark 9 (Shared random streams and Assumption 1). With a single global random stream, changing a_d can shift the order in which cross-component events consume random draws, so their realized values may differ across counterfactual actions. This affects the realized Δ_d but not $\mathbb{E}[\Delta_d \mid \mathcal{F}_d, a_d]$, so Assumption 1—which is a statement about conditional means—is unaffected. The measurements in Section 6 that fork the simulator under common random numbers exploit exactly this distinction.

All proofs are given inline where the results are stated: Proposition 1 and Proposition 3 in Section 5.2, Lemma 1 and Proposition 4 in Section 5.6, and Proposition 5 in Section 5.8. They are short enough that separating them from the statements would cost more than it saves.

Appendix B. Environments

All three environment families are Action-Evolution Petri Nets built with the same library and solved by the same agent; they differ in topology, not in machinery. Each grants reward 1 per completed task, so return is throughput over the horizon. Table B.8 collects the parameters.

Multi-site skills-based routing (the realistic environment). A firm runs four service sites. Each site has its own stochastic arrival stream of tasks of two types and its own heterogeneous pool of dedicated specialists: one skilled for each type, fast on their matched type (service delay 1) and slow otherwise (delay 3). A specialist serves a task at its own site and returns to that site’s pool. The decision is which queued task each free worker takes. Because every worker cycles within its own site, each worker’s token lineage links only decisions that used it, and the episode decomposes into eight causal components—which is why this environment measures $K = 8.0$ and a causal fan-out of exactly 1.

The environment also carries a coupling knob we do not exercise in the main results but which motivates the design: a pool of *flexible* generalists, usable at any site (delay 2, type-independent). A flexible worker that serves site i , returns to the shared pool and then serves site j links those sites’ decisions into one component. The local/flexible capacity split therefore traces the whole independence spectrum—all-dedicated gives n_{sites} components, all-pooled

Table B.8: Environment parameters. Horizon is in simulated time units; the heuristic is the anchor for the normalization $(\text{score} - \text{random}) / (\text{heuristic} - \text{random})$ and is a strong myopic rule, not a proven optimum, so normalized values above 1 are attainable.

environment	structure	horizon	random	heuristic
multi-site routing	4 sites $\times$ 2 dedicated specialists	20	62.90	80.25
N -copies, $N = 2$	2 independent copies, 1 employee each	20	15.87	22.60
N -copies, $N = 4$	4 independent copies, 1 employee each	20	31.13	44.60
N -copies, $N = 8$	8 independent copies, 1 employee each	20	63.20	89.53
single component	2 stages, 1 shared pool of 3	20	9.85	14.78

gives one—instantiating the classical OR question of resource pooling and flexibility as a single dial over K . The reported configuration is the all-dedicated end.

The N -copies scaling benchmark. N fully independent single-stage assignment problems in one net, served by one shared policy, with reward summed over copies. Each copy has its own arrival stream, queue, busy place and dedicated employee; no place is shared. A copy’s tasks are either *matched* to its employee (service $1 + U\{0, 1\}$) or *crossed* (service $3 + U\{0, 1\}$), so serving matched work first maximizes throughput. The copies are causally independent by construction, so $K \leq N$ with equality whenever every copy emits a reward within the horizon (Definition 1). This is the one environment where the independence structure is a controlled parameter rather than a measured property, which is what makes it the scaling benchmark; it is deliberately synthetic.

The single-component environment (negative control). A two-stage service line: cases arrive stochastically, are served at stage one, then queue for stage two, and reward is emitted on completing stage two. Both stages draw from *one shared pool of three heterogeneous employees*—two specialists, fast on their matched type (delay 1) and slower otherwise (2), plus a generalist (delay 3)—and every employee returns to that same pool after each service. The shared pool links every case’s lineage into a single causal component, giving $K = 1.0$ with one component holding 100% of reward mass. It is the mechanism’s own predicted null, and it is also the environment of the boundary-condition investigation of Section 7, where it is called the *bottleneck environment* because the shared pool is the binding constraint.

Heuristics. All three anchors are simple priority rules and none ever postpones. On multi-site, a specialist takes its own matched type first; failing that a flexible worker is used; failing that any specialist takes crossed work. On N -copies, a binding serving a matched (fast) task is preferred, otherwise any enabled binding is taken. On the single-component environment the rule prioritises by stage—work in progress at stage two before new work at stage one—and within a stage prefers a type-matched employee. Random baselines sample uniformly from the enabled bindings. Both anchors are evaluated on the canonical no-postpone variant of each net, so the normalization scale is a property of the environment rather than of the action space the learner sees.

Table C.9: Hyperparameters, identical for every arm and every environment.

parameter	value
algorithm	PPO with clipped surrogate objective
epochs	40 (converged budget; 15 and 30 retained as ablations)
episodes per epoch	20
minibatch size	32
policy / value learning rate	3×10^{-4} / 3×10^{-4}
policy / value update passes	3 / 4 per epoch
clip parameter ε	0.2
entropy bonus	0.01
KL early-stopping limit	0.15 (mean per-state $\text{KL}(\pi_{\text{old}} \parallel \pi)$)
discount γ / GAE λ	0.99 / 0.95
SMDP discount rate β	0.5
postponement	enabled

Appendix C. Training setup and reproducibility

Architecture.. Policy and value networks are separate heterogeneous graph networks over the assignment-graph representation, each a stack of three Heterogeneous Graph Transformer convolutions (Hu et al., 2020) with hidden width 128 and two attention heads, with residual connections and dropout. Places and transitions are distinct node types. The critic pools over node embeddings to produce a scalar; the actor decodes one logit per enabled action binding from that binding’s node embedding—the asymmetry Section 7 analyses. Input widths are inferred lazily from the first observation, so the same code runs unchanged across environments with different attribute sets.

Optimization.. All arms share the hyperparameters of Table C.9; no per-arm tuning was performed, and the causal schemes inherit the baseline’s settings unchanged. This is deliberate—any tuning advantage would confound the credit comparison—but it also means the reported numbers are not the best attainable for any arm.

Evaluation protocol.. Every second epoch the greedy (argmax) policy is evaluated on 20 episodes; the reported *final* value is the last such point and *best* the maximum over them. All evaluation uses common random numbers: with the evaluation seed fixed, evaluation episode i always runs scenario seed + i , so every method, seed and epoch is scored on one fixed scenario set and arms differ only by policy. This matters more than it may appear: without it, the measured noise floor on the single-component environment is ± 0.231 standard deviations per evaluation point and ± 0.327 on a paired difference, and drift is inflated by roughly 0.40 even for a policy that is not changing. Baselines average 40 episodes. Seeds 0–19 are used for every cell, and comparisons are paired seed-by-seed; we report paired t -tests and Wilcoxon signed-rank tests, and where they disagree we say so.

Determinism.. A single entry point seeds Python’s random, NumPy and PyTorch, and is applied to environment construction, network initialization, rollout collection and evaluation, so a cell is reproducible from its (environment, method, seed) triple. Paired arms share their network initialization. Each cell is written to its own result file and runs are resumable: re-running a sweep trains only the missing cells, which is how the arm coverage in Table 4 was filled incrementally.

Cost.. Credit computation is a single backward pass over the decisions of one episode, $O(|E|)$ in the provenance DAG, measured at approximately one millisecond per episode. Against a training cell at four copies this is under 8% of wall-clock time (PPO 7.44 minutes against 8.05); the graph network dominates, and profiling attributes roughly 83% of a cell to the optimizer’s repeated forward and backward passes over the collected graphs. No arm requires forked simulation.

Appendix D. Falsified mechanisms

Section 6.5 reports that computing the causal estimand exactly does not pay, and offers only a partial structural account of why the remaining variants order as they do. That claim would be easy to dress up with a plausible mechanism. We instead record every mechanism we tested and rejected, with the measurement that rejected it, so a reader can judge how much of the ordering is explained and how much is not. All six were pre-registered in the sense that the prediction was written into the diagnostic script before it was run.

Appendix D.1. Summary

Appendix D.2. Detail

Postpone transparency.. Under token-flow postponement a POSTPONE firing re-emits the entire marking, so it terminates the nearest-producer walk without having caused the flow, and it carries $2.25\text{--}2.66\times$ the row sum of production actions. We therefore expected removing it to make R closer to action-measurable. It does the opposite. Transparency *redirects* postpone’s edges to the true producers—0 production-to-production edges lost, 50 gained at four copies and 97 on the single-component environment—and the recovered edges are precisely the action-dependent ones. Postpone had been making R look invariant by destroying information. The resulting variant is also the worst arm we measured (0.106 normalized), so the more causally faithful graph is the worse learner—a second instance of the pattern in Section 6.

Chain decay and credit magnitude.. Both were attempts to explain a large performance gap by a mechanical property of the credit vector, and both fail on magnitude grounds: the measured differences are one to five percent where the performance difference to be explained was 0.44 in normalized return.

Table D.10: Candidate mechanisms for the ablation ordering, and the measurements that falsified them. “Inverted” means the measured quantity ordered the variants *opposite* to performance.

hypothesis	measurement	outcome
Postpone is the dominant source of the action-dependence of R	making postpone transparent leaves R identical across actions on 50.0% of fork points, down from 68.8%; spread on the single-component environment rises $0.580 \rightarrow 0.947$	worse
Credit decays along chains as $\prod w$, so CGAE-F propagates less	backward propagation mass at causal depth 5: 0.89 (CGAE-F) vs 0.95 (CGAE-CF)	too small
Larger credit magnitude gives a larger effective step	mean $ Q $ at two copies: 2.23 vs 2.22; SD 1.10 vs 1.12	identical
Per-state ranking fidelity against a measured counterfactual	Spearman ρ / top-1 agreement with common-random-number counterfactual return: CGAE-DAG +0.681 / 83.3%, CGAE-CF +0.435 / 33.3%	inverted
Credit signal-to-noise ratio (bias–variance)	between-action signal over within-action noise: CGAE-DAG 0.959 (highest), CGAE-CF 0.895; full range 0.80–0.96	inverted
CGAE-DAG’s arbitrary $\lambda^{\min}$ depth weighting	replacing it with the depth-conditioned weight $\mathbb{E}[\lambda^L \mid \text{visit}]$ moves correlation with CGAE-CF from 0.740 to 0.748	$\Delta = 0.008$

Ranking fidelity. This is the sharpest negative result, because it tests the estimator against ground truth rather than against another estimator. At a decision state we fork the simulator once per action under common random numbers and record the realized discounted return-to-go; differences across branches are then the causal effect of the action with the exogenous component cancelled. A credit scheme should rank actions the way that quantity does. On the single-component environment CGAE-DAG ranks them *better* than CGAE-CF on both measures while learning far worse. Ranking fidelity at a fixed state therefore does not determine learning outcome. Caveats: 12 usable fork points, and the continuation policy is random, so this measures counterfactual value under random continuation rather than under the converged policy.

Signal-to-noise. We forked each production action under M shared exogenous seeds and formed the ratio of between-action to within-action credit dispersion—an F -ratio for the quantity a policy gradient integrates. The prediction was that CGAE-DAG, whose closure sum approaches the unscoped return on a single-component problem, would show the lowest ratio. It shows the highest. The mechanism was half right and immaterial: CGAE-DAG does carry about twice the

noise (0.459 vs 0.272), but it also carries about twice the signal (0.420 vs 0.237), and the scaling cancels.

The depth-weighting choice. CGAE-DAG weights a descendant’s owned reward by λ^k at minimum causal depth k , which is discretionary: a re-convergent descendant reachable at depths 2 and 7 receives λ^2 . We built the principled alternative, weighting by $\mathbb{E}[\lambda^L \mid \text{the walk visits } j]$, which coincides with the closure estimand at $\lambda = 1$ and inherits CGAE-CF’s depth distribution conditioned on arrival. It changes almost nothing. The reason is structural and is worth stating as a positive by-product: since

$$c_d^{\text{CGAE-CF}}(j) = \underbrace{\Pr[\text{visit } j]}_{\text{the fan-out gap}} \cdot \underbrace{\mathbb{E}[\lambda^L \mid \text{visit } j]}_{\text{depth weighting}},$$

the two factors are separable, and the closure variants differ from CGAE-CF only by discarding the first. The behavioural difference is the gap itself, not any detail of how depth is discounted within the closure.

Appendix D.3. A budget artifact, recorded for the same reason

At the 15-epoch budget with 20 seeds, CGAE-CF measured 0.401 against CGAE-F’s 0.837 at two copies (4W/14L, $p = 0.0043$)—an apparently significant deficiency in the method we ship. It is an artifact of non-convergence. At 40 epochs the comparison is +0.102 ($p = 0.51$), sign reversed. Three measurements identify the cause: a λ sweep recovers most of the deficit ($0.401 \rightarrow 0.579 \rightarrow 0.639$ as λ falls $0.95 \rightarrow 0.85 \rightarrow 0.70$ toward CGAE-F’s effective $0.95 \times 0.896 \approx 0.851$), so it is a convergence-*speed* effect; the credit vectors differ by about one percent; and the outcomes are bimodal with nothing in between. We report this because it is the clearest evidence in this work that a significant result at a short budget can be entirely spurious, and it is why Section 6 reports the converged budget throughout.

Appendix D.4. The one surviving correlate

Across all arms and environments the single variable that separates working from failing configurations without overlap is *final policy entropy*: 0.27–0.56 for the four arms that learn, 0.13–0.15 for the three that fail. Whether this is cause or symptom is untested—a policy in a poor local optimum will show low entropy regardless of how it arrived—and the decisive experiment, an entropy-bonus intervention on the failing arms, has not been run. We state the correlation and its status rather than promoting it to a mechanism.

Appendix D.5. Methodological note

All six diagnostics share a design feature that may explain their uniform failure: each measures a *static* property of the estimator under a *random* policy with random continuations. Training involves a learned policy that becomes progressively deterministic, and the failures we are trying to explain unfold over tens of epochs. It is entirely possible that no static, random-policy property of a credit estimator predicts its training outcome. If so, that is itself a finding about how credit-assignment methods should be evaluated, and it argues for interventional rather than observational diagnostics—consistent with the boundary-condition results of Section 7, where only active-intervention mechanisms ever moved the single-component floor.

Appendix E. Boundary-condition investigation: technical detail

Protocol, and why it differs.. The three experiments of Section 7 predate the converged-budget re-runs and use the stochastic suite’s standard budget: the single-component environment at 30 epochs $\times$ 20 episodes, 10 seeds, paired seed-by-seed against a ppo_clip baseline, reporting normalized final return. That baseline therefore reads 0.739 ± 0.10 here against 0.729 ± 0.12 in Table 6, which is the same arm at 40 epochs and 20 seeds; the difference is budget and seed count, not environment. We did not re-run them at the longer budget because all three outcomes are nulls or negatives, and a longer budget cannot convert a null into the positive result that would change the section’s conclusion. The environment is the one described in Appendix Appendix B; Section 7 calls it the *bottleneck environment* to name the mechanism under test rather than the statistic K .

Potential-based shaping.. The shaping term is the SMDP generalization $F(s, a, s') = e^{-\beta\tau}\Phi(s') - \Phi(s)$, added to every step’s reward, with the potential read off the net’s topology: $\Phi(s) = \sum_p \text{decay}^{\text{hop}(p)} \cdot \min(|p|, \text{cap})$, a distance-decayed structural backlog in which each place’s token count is weighted by its hop-distance to the nearest reward-emitting transition and optionally capped. Policy invariance (Ng et al., 1999) was verified in code by an exact terminal-value cancellation test rather than merely cited. The coefficient dial reuses one cached baseline across all four points, so the comparison is paired by construction.

The failure mechanism is worth stating because it is not a property of the potential’s shape. This environment has constant exogenous arrivals—one case per time unit—so the backlog potential rises as often as it falls, and the per-step shaping signal is of the same order of magnitude as the entire episode return. Aggregated over an episode the term telescopes to a constant, exactly as the theorem promises, but at any finite budget it is a substantially noisier value-regression target. Safety and efficacy are independent properties, and this mechanism has the first without the second.

Structural input features.. Two feature sets were appended to each action node’s own inputs: first, whether an action type structurally competes for a token and with how many rivals, derived from the conflict graph; then, after that returned a null, the minimum topological hop-distance from an action’s required inputs to the nearest reward-emitting transition. The second does discriminate between the environment’s two action types (0.656 against 0.810), so the null is not a degenerate-feature artifact.

The redundancy argument is the stronger result and does not depend on either feature’s design. Every action node already carries a one-hot action-type identity at zero hops. Any scalar that is a function of action type alone is therefore a deterministic function of information a single learned weight on that one-hot could reconstruct, so it cannot add representational capacity. This holds for this architecture on any topology, which is why we report it as a proof rather than as two failed experiments. A structural feature can only help if it varies *within* an action type—across markings—which conflict-degree on a two-node conflict pair provably does not.

Actor architecture. The blind spot is a property of the computational graph, not of training. The actor decodes each action’s logit from that action node’s own final message-passing embedding; the graph’s edges run strictly along token flow, with no reverse edges at any real observation call site; so an action’s logit can depend on state only through a forward path reaching it within the network’s depth. The demonstration uses a purpose-built topology of two structurally independent two-stage pipelines: perturbing one pipeline’s downstream marking leaves the other pipeline’s action logit bit-for-bit unchanged ($\Delta = 0.000$) under a randomly initialized, untrained network. Reachability does not depend on learned weights, so no training budget can close it.

The fix concatenates onto each action node, before decoding, the same pooled context the critic already computes—an intervention on the actor alone, leaving the critic and the credit path untouched. On the proof topology it closes the gap exactly ($\Delta = 0.000 \rightarrow 0.528$). On the single-component environment it is neutral (0.746 ± 0.14 against 0.739 ± 0.10 , $p = .91$), and the two results are not in tension: that environment’s shared resource pool recycles workers, which gives the unfixed architecture a marginal, accidental path to the same information within three hops. The blind spot was never total there, and whatever fraction the fix newly exposes is not decision-relevant enough to move a well-tuned baseline at this budget. The architectural finding stands on the proof; the training result bounds its practical value on this environment only.

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